



Kalyakin, Ivan. 2026. VP ellipsis and argument structure alternations: Evidence from Muira Dargwa complex predicates. *Glossa: a journal of general linguistics* 11(1). pp. 1–47. DOI: <https://doi.org/10.16995/glossa.20119>



VP ellipsis and argument structure alternations: Evidence from Muira Dargwa complex predicates

Ivan Kalyakin, Institute of Linguistics, Russian Academy of Sciences; Pushkin State Russian Language Institute,
kalyakin.iv@gmail.com

In this paper, I investigate the properties of an elliptical construction in the Nakh-Daghestanian language Muira Dargwa, in which only part of a complex predicate goes missing. I show that this process has a number of common features with Verb Phrase Ellipsis in English and related phenomena in other languages. Under an analysis of complex predicates where the light verb is the realization of the *v* head, the deleted constituent can be associated with the complement of *v*. Thus, the process diverges from similar phenomena in that the size of the ellipsis is smaller, allowing for a larger number of mismatches between antecedent and elliptical clauses.

Glossa: a journal of general linguistics is a peer-reviewed open access journal published by the Open Library of Humanities. © 2026 The Author(s). This is an open-access article distributed under the terms of the Creative Commons Attribution 4.0 International License (CC-BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See <http://creativecommons.org/licenses/by/4.0/>.

OPEN ACCESS



1 Introduction

Verb Phrase Ellipsis (VPE) is a process in which a verb phrase, often identified as *vP* (Aelbrecht 2010; Merchant 2013c), goes missing under identity with the *vP* of an antecedent clause. An example of VPE in English is given in (1), where the antecedent is in bold and Δ indicates the elided phrase.

- (1) John **bought a present for Meg**, and Fred did Δ , too. (Δ = [*vP* buy a present for Meg])

Although VPE had long been considered to be unique to English, research over the past few decades has shown that VPE may also exist in other languages as well, albeit in a slightly different guise. In verb-raising languages (such as Russian, Greek, Hindi-Urdu, Lithuanian, Portuguese, Swahili), the verb manages to escape the ellipsis site before the deletion occurs; such a variety of ellipsis has been labeled as Verb-Stranding Verb Phrase Ellipsis (VVPE), see (Goldberg 2005) for an extensive study of the phenomenon. In (2) from Russian, only the internal arguments go missing, as the main verb raises to a higher functional projection (presumably, *Asp*), before *vP* is elided.¹

- (2) *Russian* (Gribanova 2013: 106)

Dina	kupila	svoej	dočke	škol'nye	učebniki,	a
Dina.NOM	bought.SG.F	self.DAT	daughter.DAT	school.ACC	textbooks.ACC	but
Paša	ne	kupil	—	—		
Paša.NOM	NEG	bought.SG.M				

‘Dina bought her daughter textbooks, but Paša didn’t.’

The existence of VPE in these languages suggests that it may occur yet in other languages as well, though the phenomenon may be obscured by some other factors, as was the case with head movement. Moreover, it has been demonstrated that VVPE is not a uniform phenomenon, either across different languages or across different constructions within a single language: the process may target constituents of different size (Lipták 2012), Verbal Identity Requirement (Goldberg 2005) violations may be tolerated or not, etc. However, since (Otani & Whitman 1991), the common wisdom shared by virtually all researchers is that the existence of VVPE presupposes head movement, which allows the verb to escape deletion. So, another option for getting a VVPE-like string, which has not been thoroughly investigated yet, is that the verbal element is “stranded”

¹ In a series of recent papers, Landau (2020a;b) argues against a VVPE analysis of object gap sentences in Russian and in verb-raising languages in general, see (Kasenov & Rudnev 2025) and references therein for additional arguments against VVPE in Russian. However, the debate about the universal unavailability of VVPE is far from over, see (Manetta 2021; Simpson 2023).

not because it manages to evacuate the ellipsis site via head movement, but rather because the constituent that is deleted is smaller than that hosting the verbal element.

In this paper, I examine an ellipsis construction attested in the Nakh-Daghestanian language Muira Dargwa, in which part of a complex predicate (CPr) goes missing, and argue that, unlike previously examined languages, ellipsis in Muira Dargwa targets a constituent smaller than *vP*. In the example of the construction in (3), not only the internal argument *χureg* ‘food’ is deleted, but also the non-verbal element (NV) *wana* ‘warm’ of the CPr *wana* AGR-*arq*’- ‘to warm smth. up’.²

- (3) Pat’imat-li χureg wana b-arq’-ib, Ashura-li = ra (χureg wana)
 Patimat-ERG food.ABS warm N-make.PFV-AOR Ashura-ERG = ADD food.ABS warm
 b-arq’-ib
 N-make.PFV-AOR
 ‘Patimat warmed up the food, and Ashura did too.’

A similar construction has been attested in Persian and first described in (Toosarvandani 2009), where it got labeled as *v*-Stranding Verb Phrase Ellipsis (*vVPE*), though see (Shafiei 2015; Rasekhi 2016; 2018; Ahari 2024) for critical discussion and alternative proposals. I suggest that *vVPE* also exists in Muira Dargwa, with (3) being an instantiation of the construction. As noted in (Erschler 2021), ellipsis in CPrs in Nakh-Daghestanian languages, though it superficially resembles *vVPE* in Persian, has not been previously described and requires investigation. So, my first aim is to provide the data from Muira Dargwa to the linguistic community, demonstrating further evidence in favor of the need to extend the typology of ellipsis constructions in the languages of the world.

My second aim is to develop an analysis of *vVPE* in Muira Dargwa and to show that the construction, similarly to both *VPE* in English and *VVPE* in verb-raising languages, is subject to well-known requirements (Lobeck 1995; Merchant 2001): (i) it must be properly licensed and (ii) it must obey an antecedence condition. In addition to the similarities, I discuss some aspects in which Muira Dargwa *vVPE* differs from its better-studied relatives. The differences, I argue, stem from the size of the elided constituent, which in *vVPE* appears to be smaller than that in (*V*)*VPE*. So, the data presented in this paper provide novel evidence in favor of the line of theories that attribute the (im)possibilities of featural mismatches across different types of ellipsis to the size of the elided constituent (Merchant 2013c; Sailor 2014).

The paper is structured as follows. First, in Section 2, I give background information on Dargwa languages and discuss the structure of CPrs. In Section 3, I present rich evidence that the gap in examples like (3) is derived via ellipsis and cannot be a null pro-form. Having established this, it is still required to differentiate between two possible analyses of the construction: *vVPE*, which deletes everything within the ellipsis site in one fell swoop, and Argument Ellipsis, a

² Throughout the paper, the elided constituent will be indicated with angle brackets.

more targeted operation that achieves the same result by deleting elements one by one. I then demonstrate that the former is the right analysis, though the latter seems to be also operative in Muira Dargwa. Next, Section 4 addresses the two main ingredients of ellipsis licensing. In Section 5, I investigate the interaction between ν VPE and a number of argument structure phenomena – inchoative-causative alternations, causativization, and antipassivization – and discuss the ramifications for existing theories of ellipsis. Then, Section 6 provides a bird’s eye view of ν VPE in some other languages, noting some available parameters of variation. Finally, Section 7 concludes.

2 Dargwa languages

In this section, I provide the most basic background information on Dargwa languages (§2.1). For a more complete picture, see (Ganenkov & Maisak 2021) on the Nakh-Daghestanian language family and (Sumbatova 2021) on Dargwa particularly. Then, I proceed to discuss the formation of CPRs (§2.2).

2.1 General background

Dargwa languages constitute a separate branch of the Nakh-Daghestanian language family and are spoken primarily in the central part of the Republic of Dagestan (Russian Federation). It has been traditionally assumed that different varieties of Dargwa constitute dialects of a single language. Nowadays, however, there is a consensus that it is necessary to distinguish between several Dargwa languages, the number of which ranges from six (Mutalov 2021) to fifteen (Koryakov 2021). According to Koryakov (2021), Muira Dargwa, the language under discussion, belongs to the north-central group and is one of the largest languages of the branch, spoken by approximately 40.000 people.

As for linguistic characteristics, all Dargwa languages are left-branching with SOV as the unmarked word order; it should be noted, however, that the word order enjoys a lot of freedom (Forker 2021). The languages allow for extensive *pro*-drop, employing null pronominals for direct objects as often as for subjects; null realization of indirect objects is also attested, although it is a much less frequent phenomenon. Typically for Nakh-Daghestanian languages, Dargwa languages are morphologically ergative, both in terms of case and (gender) agreement, and syntactically accusative. They are characterized by rich agglutinating morphology, both in the nominal and verbal domains.

The data in this article come from author’s own field work with a dialect of Muira Dargwa spoken in the village of Kalkni (Dakhadaevski District, Republic of Dagestan, Russia). All acceptability judgments were elicited between 2022 and 2023 from at least three different native speakers of the language, aged between 30 and 65, and subsequently confirmed in different sessions.

2.2 Complex predicates

In Dargwa languages, the verbal lexicon is traditionally divided into three classes of predicates:

- i. Simplex verbs
- ii. Preverb verbs
- iii. Complex verbs (*aka* CPrs)

The class of simplex verbs is closed and numbers no more than 200 stems. These stems have a very simple phonological structure, in most cases consisting of two or three phonemes; some of them imply the presence of a gender agreement prefix. Absolute majority of underived stems in Dargwa languages come in pairs that express the aspectual opposition between perfective and imperfective. Due to the paucity of verbs in this class, most of Dargwa verbs are either CPrs or derived via the addition of spatial prefixes (preverbs) to simple stems. The formation of both preverb and complex verbs is highly productive, the two classes comprise an ever expanding segment of the verbal lexicon.

CPrs are comprised of two parts: a light verb (LV) and a non-verbal element (NV). LVs have rather bleached semantics (causation, initiation, etc.), function as verbalizers and play a crucial role in determining the argument structure of the resulting CPrs. Most of the LVs are homophonous with simplex verbs, which bear full lexical meaning – these are traditionally called “heavy verbs”. In comparison to other languages of the Caucasus and the world, the number of LVs in Muira Dargwa is quite high – so far, no less than 20 LVs have been encountered (Kalyakin 2023). However, most of them have rather limited distribution, combining with just a few (or even one) NVs. The list of the most commonly used LVs glossed with their heavy version interpretation is presented in (4).³ In what follows, I will primarily focus on CPrs with LVs *-arq'* / *-irq'* ‘make, do’ and *-Ø* / *-irh* ‘be, become’.

- (4)
- | | |
|-----------------------------|---------------------|
| <i>-arq'</i> / <i>-irq'</i> | ‘make, do’ |
| <i>-Ø</i> / <i>-irh</i> | ‘be, become’ |
| <i>-uq</i> / <i>-ulq</i> | ‘move’ |
| <i>-ik'</i> | ‘speak’ (IPFV only) |

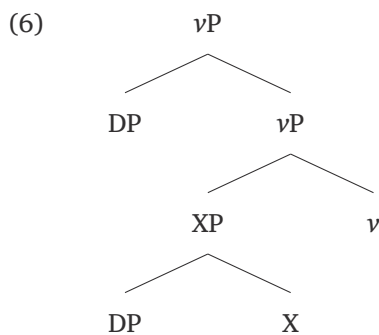
NVs are usually represented by nominal and adjectival roots, as well as ideophones, adverbs, numerals and borrowed verbs (primarily from Arabic, Persian and Russian), which are also used as NVs. The main function of an NV is to determine the lexical meaning of a CPr. Examples of CPrs with different types of NVs are provided in (5).

³ The first verb in the pair is perfective, and the second is imperfective. Hyphen before verb root indicates that the root implies the presence of a gender marker.

- (5)
- a. *Noun*
gap ‘praise’ + AGR-*arq*’- ‘do’ = ‘to praise’
taman ‘finish’ + AGR-*i*- ‘become’ = ‘finish’
 - b. *Adjective*
šinix ‘green’ + AGR-*irh*- ‘become’ = ‘become green’
ʔaʕh ‘good’ + AGR-*irh*- ‘become’ = ‘recover’
 - c. *Ideophone*
maʕw ‘meow’ + *at*’- ‘say’ = ‘meow (once)’
 - d. *Adverb*
cahna ‘together’ + AGR-*ik*- ‘fall’ = ‘get together’
 - e. *Numeral*
čʷal ‘two’ + AGR-*arq*’- ‘make’ = ‘split in half’
 - f. *Borrowed verb*
podderžiwatʕ ‘support (Russ.)’ + AGR-*arq*’- ‘do’ = ‘support’

Now we can turn to the structure of CPrs. Many syntactic accounts of CPr formation have productively employed a decomposed verbal structure à la (Hale & Keyser 2002). Despite the wide variety of differences, these accounts share the notion that the LV originates at a point in structure higher than the NV, see (Folli et al. 2005; Megerdooomian 2011) for Persian, (Manetta 2018; 2019) for Hindi-Urdu, and (Tabak & Demirok 2024) for Turkish.

One of the most influential analyses was developed in (Folli et al. 2005) for Persian. Under this account, argument structure of a CPr is dependent on both the LV and NV: the former determines whether an external argument will be projected, while the latter selects for the internal argument(s). On the basis of this division of labor, Folli et al. (2005) suggest that the structure in (6), where both the external and internal arguments are projected, underlies transitive (causative) CPrs.



An intransitive (inchoative) CPr is formed by substituting a transitive LV with a corresponding intransitive one. A minimal pair is presented in (7): the LV in (7a) selects for an external argument, while the LV in (7b) does not. Similar examples from Muira Dargwa are shown in (8).

(7) *Persian* (Folli et al. 2005: 1376)

- a. Minu bachcha-ro kotak zad
Minu child-RA beating hit
'Minu hit the child.'
- b. Bachche kotak xord
child beating collided
'The child got hit.'

- (8) a. Pat'imat-li šin wana d-arq'-ib
Patimat-ERG water.ABS warm NPL-make.PFV-AOR
'Patimat warmed up the water'
- b. Šin wana d-Ø-ib
water.ABS warm NPL-become.PFV-AOR
'The water has warmed up.'

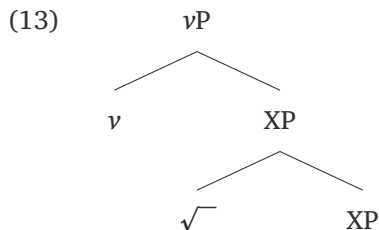
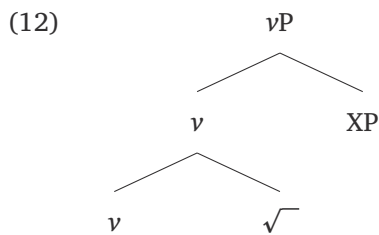
Due to a number of similarities between Persian and Muira Dargwa CPrs, I partially extend the analysis of Folli et al. (2005) to Muira Dargwa, treating the LV as an overt realization of the *v* head. I depart from their analysis in a number of important details, however. First, I take the NV to be the realization of a bare root (Marantz 1997) that corresponds to a phonological word. This explains the "deficient" morphosyntactic nature of NVs, which can be observed in a number of respects (Kalyakin 2023; 2025b). Most importantly, despite the fact that "nominal" NVs superficially resemble bare noun stems and that absolutive case is zero-marked, these NVs bear no case and are invisible to case calculus. This effectively explains the presence of another zero-marked nominal, which actually bears absolutive case, and the inability of these NVs to participate in the agreement relation – as (9) shows, the NV *gap* 'praise' is not in absolutive, since there is another zero-marked DP *Rasul* with which the verb agrees in gender and number. Besides that, the element cannot host modifiers (10) and it is not referential (11).

- (9) Učitiġ-li Rasul gap *d-arq'-ib / w-arq'-ib
teacher-ERG Rasul.ABS praise NPL-make.PFV-AOR M-make.PFV-AOR
'The teacher praised Rasul.'

- (10) *Darĥa [qa^hrq-il duc'] Ø-ik'-u-r = ri
child.ABS quick-ATR run M-say.IPFV-PROG-CVB = PST
'(The child ran fast.)'

- (11) *Murad [iř taʰh] Ø-uq-un
 Murad this.SP.ABS jump M-move.PFV-AOR
 ('Murad made that jump.')

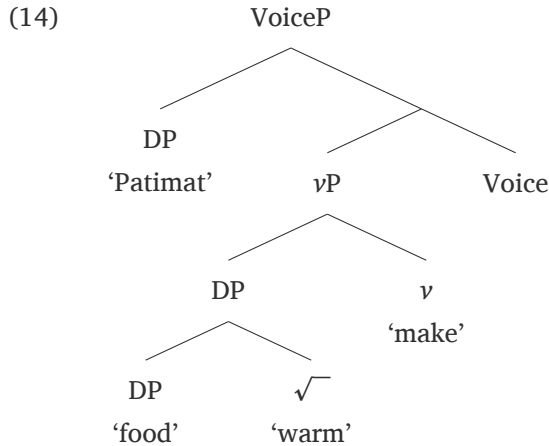
Furthermore, I adopt the now widely accepted view that roots are devoid of syntactic information, viz. they do not project and do not subcategorize (Alexiadou 2014; De Belder & van Craenenbroeck 2015; Merchant 2019). With respect to the position in the syntactic structure, roots can be attached to v (12), or merged with the direct object (13) – the exact position is determined by the root's conceptual content. The intuition here is that the roots that modify the event introduced by v , for example imposing a manner on an activity event ($\sqrt{\text{RUN}}$, $\sqrt{\text{HAMMER}}$), are attached directly to v , while roots that describe the end state that is associated with a direct object of a change-of-state event ($\sqrt{\text{BREAK}}$, $\sqrt{\text{OPEN}}$) are attached to the direct object DP (Marantz 2009a;b; 2013; Irwin 2012; Levinson 2014). This dichotomy will prove important for the current study of v VPE, see §4.1.



Finally, concerning the structure of the verbal domain, I follow (Pylkkänen 2008; Harley 2013) and postulate two functional heads, Voice and v , which share functions that are usually associated with a single head, e.g. v in (Chomsky 1995). Voice is endowed with the ability to host external arguments in its specifier, such as agents or causes, while v functions as a categorizer and introduces eventive semantics.

Given all this, the structure of a causative change-of-state CPr, e.g. *wana* AGR-*arq'*- 'to warm smth. up' in (3), will look as in (14). As can be seen, the structure here is an extended version of the tree (13); I will return to the semantic interpretation of this structure in §4.2. The inchoative counterpart of this CPr will have the same structure, except that the Voice head disallows

merging of a specifier, viz. it does not introduce an external argument and does not bear any thematic information.



3 Diagnosing ellipsis

It has long been argued that there is a distinction between pronouns and ellipsis, or *deep* and *surface anaphora* in terms of Hankamer & Sag (1976). Pronouns are considered to not have an articulate internal structure, while ellipsis, by virtue of being a surface phenomenon, presupposes the existence of syntactic structure, which, however, is left unpronounced.

So, the first thing to do is demonstrate that the sentences like (3) do exemplify surface anaphora. In this section, I use several diagnostics that have been proposed in the literature to provide evidence that the elements are missing not due to deep anaphora. The diagnostics are: pragmatic control (§3.1), variable binding (§3.2), scope phenomena (§3.3), extraction (§3.4), and agreement (§3.5). Furthermore, even after establishing that we are dealing with ellipsis, we must still demonstrate that the process responsible for non-pronunciation is indeed *vVPE*, as there are other sub-types of ellipsis which, in principle, are capable of generating *vVPE*-like strings. So, before moving on to developing an analysis, in §3.6, I show that the relevant construction must be derived via *vVPE* and not Argument Ellipsis, yet another – more targeted – mechanism of non-pronunciation of syntactic material.

3.1 Pragmatic control

In their seminal work, Hankamer & Sag (1976) observe that ellipsis must have a linguistic (syntactic) antecedent rather than a merely contextual one. In other words, ellipsis does not allow what Hankamer & Sag call *pragmatic control*, see (15a). On the other hand, interpretation of deep anaphora depends on the particular discourse model, so pragmatic control is possible, as illustrated in (15b).

(15) (Hankamer & Sag 1976: 392)

[Context: Sag produces a cleaver and prepares to hack off his left hand]

Hankamer:

- a. #Don't be alarmed, ladies and gentlemen, we've rehearsed this act several times, and he never actually does Δ .
- b. Don't be alarmed, ladies and gentlemen, we've rehearsed this act several times, and he never actually does it.

Similarly to English VPE, Muira Dargwa vVPE prohibits pragmatic control. Example (16a) shows that despite the presence of a non-linguistic antecedent, it is impossible to delete the internal argument *χureg* 'food' along with the NV *wana* 'warm' of the CPR *wana* AGR-*arq*'- 'to warm smth. up'. Note that the omission of only the internal argument is possible (16b), so it is reasonable to assume that in this case we are dealing with deep anaphora, though the pronoun has no phonological realization.

(16) [Context: There is a plate of food on the table, in front of which a child sits and raises his hands over it, checking whether his mother has warmed it up.]

Me to the child:

- a. #Aba-li ⟨χureg wana⟩ b-arq'-ib
 mother-ERG food.ABS warm N-make.PFV-AOR
 ('You can eat now.) The mother has warmed the food up.')
- b. Aba-li (il) wana b-arq'-ib
 mother-ERG this.ADDR.ABS warm N-make.PFV-AOR
 ('You can eat now.) The mother has warmed it up.'

An additional example with the CPR *gap* AGR-*arq*'- 'to praise' is provided in (17).

(17) [Context: After the class, the teacher praised Rasul, but didn't praise Murad, who also worked well.]

Me to Murad:

- #Učitiĭ-li ⟨ħu gap⟩ w-arq'-at:i
 teacher-ERG you.SG.ABS praise M-make.PFV-2SG.AOR
 ('Don't be sad, Murad!) The teacher praised you. (You just didn't notice.)')

The relevance of the diagnostic to differentiate between surface and deep anaphora has been challenged in literature, see (Schachter 1977; Merchant 2005; Weir 2014) for critical discussion. In some cases, surface anaphora may be felicitous without a linguistic antecedent, and deep anaphora may fail to find a contextual antecedent. Nevertheless, so far all the judgments I have

obtained show quite a strong contrast. Moreover, I have encountered no instances of vVPE without a linguistic antecedent.

3.2 Bound variables

Johnson (2001) points out that anaphoric elements in the antecedent and ellipsis clauses can be interpreted by two different mechanisms – *coreference* and *binding*. Under coreference, *strict identity* emerges – pronouns pick the same referent in the antecedent or refer to some other individual in the discourse. Binding, in turn, interprets each pronoun as a variable, which acquires its value from a c-commanding DP (usually, subject) in the local domain (usually, clause). Binding is said to produce *sloppy identity* readings.

English VPE is known for permitting both sloppy and strict identity readings, as exemplified in (18), see (Kehler 2002) for discussion and important qualifications. That is, the sentence does not commit one to thinking that John slapped Fred; it may well be the case that John slapped himself. On the other hand, deep anaphora, having no complex internal syntactic structure and thus no elements to be interpreted as variables, is unable to provide sloppy identity readings.

- (18) Fred slapped himself because John did Δ . (Strict – OK; Sloppy – OK)

Muira Dargwa vVPE aligns with English VPE once again. Examples (19) and (20) demonstrate that both strict and sloppy readings are readily available.⁴

- (19) Asijat-li sun-na aba razi r-arq'-ib, Sabijat-li = ra
 Asiyat-ERG self-GEN mother.ABS glad F-make.PFV-AOR Sabiyat-ERG = ADD
 ⟨sun-na aba razi⟩ r-arq'-ib
 self-GEN mother.ABS glad F-make.PFV-AOR
 'Asiyat made her mother happy, and Sabiyat did too.' (Strict – OK; Sloppy – OK)

- (20) Murad-li sun-na k:urti q:a^r a-b-ert-ib, Musa-li ⟨sun-na k:urti
 Murad-ERG self-GEN shirt.ABS tear UP-N-LV.PFV-AOR Musa-ERG self-GEN shirt.ABS
 q:a^r⟩ h-e-b-ert-ib
 tear NEG-UP-N-tear.PFV-AOR
 'Murad tore his shirt, but Musa didn't.' (Strict – ?; Sloppy – OK)

Like virtually all languages of the Nakh-Daghestanian family, Muira Dargwa has no dedicated third person pronouns. Instead, a number of demonstrative pronouns are employed. Importantly, as (21) shows for the demonstrative *il* 'this, close to the addressee', these pronouns prohibit sloppy identity readings. Given the contrast between (19)–(20) and (21), it is reasonable to conclude that the former examples are derived via a different mechanism, namely, ellipsis.

⁴ At this point, I have no explanation for the degraded acceptability of strict identity reading in example (20).

- (21) Asijat-li sun-na aba razi r-arq'-ib, Sabijat-li = ra
 Asiyat-ERG self-GEN mother.ABS glad F-make.PFV-AOR Sabiyat-ERG = ADD
 il razi r-arq'-ib
 this.ADDR.ABS glad F-make.PFV-AOR
 'Asiyat made her mother happy, and Sabiyat made her happy too.'
 (Strict – OK; Sloppy – *)

As was the case with pragmatic control, the availability of sloppy identity readings has been criticized for not being exclusive to ellipsis (Hoji 1998; Merchant 2013a). So, even though the diagnostic seems to be useful at the moment, from a cross-linguistic perspective, its reliability should be taken with a grain of salt.

3.3 Inverse scope

Quantification elements inside the ellipsis site can take wide scope over the operators outside the elided constituent. Such inverse scope readings are expected if we deal with quantifier raising out of a regular, though unpronounced, syntactic structure. On the other hand, pronominal anaphors, being an instance of surface anaphora, are not expected to allow inverse scope, cf. (22).

- (22) A doctor examined every patient,
 a. ...and then a nurse did Δ . ($\exists \gg \forall$; $\forall \gg \exists$)
 b. ...and then a nurse did it. ($\exists \gg \forall$; * $\forall \gg \exists$)

As can be seen from (23) and (24), which are parallel to (22a), Muira Dargwa vVPE mirrors English VPE by allowing for both surface and inverse scope readings.

- (23) Har tuχtur-li zaʔipil ʔaʕh w-irq'-u, wa har medsestra-li = ra
 each doctor-ERG patient.ABS good M-make.IPFV-TH and each nurse-ERG = ADD
 <zaʔipil ʔaʕh> w-irq'-u
 patient.ABS good M-make.IPFV-TH
 'Every doctor treats a patient, and every nurse does too.' ($\exists \gg \forall$; $\forall \gg \exists$)
- (24) Aba-li haʕbal darha parʔat b-arq'-ib, at:a-li = ra (haʕbal
 mother-ERG three child.ABS calm HPL-make.PFV-AOR father-ERG = ADD three
 darha parʔat) b-arq'-ib
 child.ABS calm HPL-make.PFV-AOR
 'A mother calmed three children, and a father did too.' ($\exists \gg 3$; $3 \gg \exists$)

3.4 Extraction

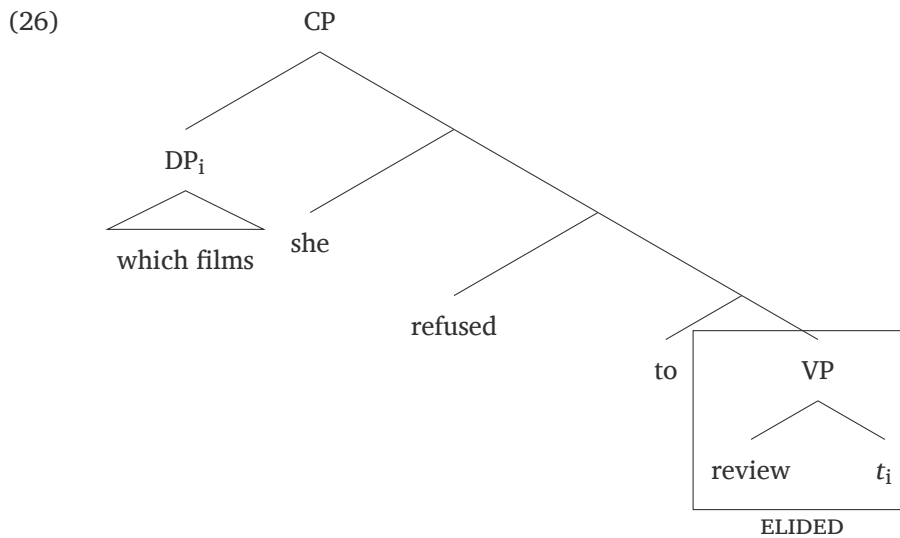
Extraction seems to be one of the most reliable diagnostics that help identify an active, though invisible, syntactic structure. The logic behind this diagnostic is very simple: if an ellipsis site

contains internal syntactic structure, it should be possible to extract something from that structure before the ellipsis occurs, thus saving the moved constituent from being deleted. All types of extraction from ellipsis sites – A- (Aelbrecht 2010), A'- (Schuyler 2001), and X⁰-extraction (Goldberg 2005) – are attested and discussed in the literature.

Example (25) illustrates A'-movement out of the elided VP. As can be seen in the structure of the second clause in (26), the *wh*-phrase is base-generated as an internal argument of the verb and later displaced to the left periphery of the clause, vacating the ellipsis site.

(25) (Merchant 2018: 248)

We need to know which films Anna agreed to review, and which ones she refused to Δ.



In turn, Null Complement Anaphora, providing no position to move out from, fails to license the extraction of the object of an understood complement predicate, cf. (27). The same goes for pronominal anaphors like *do it* in English, cf. (28).

(27) (Merchant 2018: 248)

- a. We asked Anna to review these five films, and she agreed. (*sc.* to review them)
- b. *We need to know which films Anna refused to review, and which ones she agreed.

(28) a. Meg will eat pancakes, but TOMATOES, I know she won't Δ.

- b. *Meg will eat pancakes, but TOMATOES, I know she won't do it.

Example (29) from Muira Dargwa demonstrates that constituents can escape ellipsis by undergoing rightward movement to the postverbal position (this movement is driven by information structure purposes, see Forker 2021).⁵ In such cases, *v*VPE starts to resemble

⁵ I remain agnostic to a particular analysis of the phenomenon in this case.

pseudogapping in English, which is often analyzed as a subtype of VPE (Jayaseelan 1990), but see (Johnson 2009) for a dissenting view.

- (29) Učitol-li Musa parvat w-arq'-b = ak:u, *pro* <*t*_i parvat>
 teacher-ERG Musa.ABS calm M-make.PFV-AOR = NEG.COP calm
 w-arq'-ib ʔa^hbdulla_i
 M-make.PFV-AOR Abdullah.ABS
 'The teacher didn't calm Musa down, but ABDULLAH, he did.'

A'-movement can also target the position somewhere above the verbal domain (30a). If the internal argument is not contrastive focus-marked, then there is nothing to sanction this movement and *v*VPE associated with the pseudogapping-like derivation fails, as (30b) exemplifies. This fact will be essential in proving that the process responsible for the non-pronunciation is *v*VPE, I will return to it in §3.6.

- (30) a. Učitol-li Musa parvat w-arq'-ib, *pro* Pat'imat_i <*t*_i parvat>
 teacher-ERG Musa.ABS calm M-make.PFV-AOR Patimat.ABS calm
 r-arq'-b = ak:u
 F-make.PFV-AOR = NEG.COP
 'The teacher calmed Musa down, but PATIMAT, he didn't.'
- b. *Učitol-li Musa parvat w-arq'-ib, direktur-li = ra Musa_i
 teacher-ERG Musa.ABS calm M-make.PFV-AOR principal-ERG = ADD Musa.ABS
 <*t*_i parvat> w-arq'-ib
 calm M-make.PFV-AOR
 ('The teacher calmed Musa down, and the principal did too.')

Muira Dargwa *wh*-phrases, similarly to their English counterparts, can be moved out of the silent constituent, cf. (31).

- (31) Nu-ni b-alh-u-li ak:wasi Ibrahim-li kut:i darha ʔa^hh
 I-ERG N-know.IPFV-PROG-CVB NEG.COP.1 Ibrahim-ERG which child.ABS good
 h-Ø-arq'-ib = al, amma nu-ni b-alh-u-r = ra kut:i_i *pro*
 NEG-M-make.PFV-AOR = IQ but I-ERG N-know.IPFV-PROG-CVB = 1 which.ABS
 <*t*_i ʔa^hh> w-arq'-ib = al
 good M-make.PFV-AOR = IQ
 'I don't know which child Ibrahim didn't cure, but I know which one he DID cure.'

Finally, A-movement out of the ellipsis site is also permitted, as (32) shows. Some additional examples of this type of movement will be presented in §5.

- (32) ʔinc-bi hint'in $\text{d-}\emptyset\text{-ib}$, $\text{pamidur-ti} = \text{ra}_i$ $\langle t_i \text{ hint'in} \rangle$
 apple-PL.ABS red NPL-become.PFV-AOR tomato-PL.ABS = ADD red
 $\text{d-}\emptyset\text{-ib}$
 NPL-become.PFV-AOR
 ‘The apples turned red, and the tomatoes did too.’

3.5 Agreement

All the examples presented so far contain additional evidence for surface anaphora – agreement. Since ellipsis is an operation that removes only phonological feature matrices of terminals, while their formal feature matrices are left untouched, elements inside the ellipsis site can participate in formal operations that follow the process of deletion, though see (Aelbrecht 2010; Baltin 2012). Examples (29), (30a) and (32) above show that LVs agree with missing internal arguments in gender and number. Person agreement is also available (33), see also (17).

- (33) At:aba-li nu ummaj $\text{w-arq'ib} = \text{ra}$, $\text{at:at:a-li} = \text{ra}$
 grandmother-ERG I.ABS kiss M-make.PFV-AOR = 1 grandfather-ERG = ADD
 $\langle \text{nu} \text{ ummaj} \rangle \text{ w-arq'ib} = \text{ra}$
 I.ABS kiss M-make.PFV-AOR = 1
 ‘The grandmother kissed me, and the grandfather did too.’

3.6 Against alternative analyses

In the previous subsections, we saw that all diagnostics converge on the fact that clauses missing both an internal argument and an NV are derived via ellipsis, or, in other words, they instantiate surface anaphora. However, as the discussed properties are not limited exclusively to ν VPE, it is worth exploring whether the phenomenon we are dealing with is really ν VPE, rather than, for example, Argument Ellipsis (AE), a more targeted ellipsis process which can elide syntactically fully present individual arguments independently. For instance, Hoji (1998) shows quite convincingly that what appear to be sloppy identity readings are not definitively characteristic of VPE in Japanese. Moreover, the very existence of VVPE, a phenomenon which seems to be quite similar to ν VPE, has recently attracted a lot of skepticism (Landau 2020a;b).

Returning to the baseline example, the second clause of (3), repeated here as (34), could in principle have one of three possible derivations represented in (35).⁶ It has already been

⁶ It could be additionally assumed that an NV can also be silent due to *pro*-drop or AE. I believe, however, that it is unlikely, since *pro*-drop is a kind of deep anaphora that is dependent on some salient discourse referent, and, secondly, because *pro* is a pronominal, viz. a DP. As was discussed in §2.2, an NV is not a DP, but rather a head. Similarly, AE cannot be behind the non-pronunciation of an NV, since it elides (argument) phrases, not heads. Finally, it should be noted that analyses (35b) and (35c) involve “eliding” a root individually, whatever the term might imply in that kind of a scenario. To my knowledge, a process like this one is unattested. These scenarios are considered solely for the sake of completeness, in order to identify ν VPE with greater certainty.

established that (35c) cannot be the correct analysis. Now, it remains to provide evidence that (35b), where the two elements undergo deletion separately, is also on the wrong track, thus leaving (35a) as the only right way.

- (34) Pat'imat-li χureg wana b-arq'-ib, Ashura-li = ra ⟨χureg wana⟩
 Patimat-ERG food.ABS warm N-make.PFV-AOR Ashura-ERG = ADD food.ABS warm
 b-arq'-ib
 N-make.PFV-AOR
 'Patimat warmed up the food, and Ashura did too.'
- (35) a. *Analysis 1: vVPE*
 Ashura-ERG ⟨food.ABS warm⟩ N-make.PFV-AOR
 b. *Analysis 2: AE*
 Ashura-ERG ⟨food.ABS⟩ ⟨warm⟩ N-make.PFV-AOR
 c. *Analysis 3: pro-drop*
 Ashura-ERG *pro* ⟨warm⟩ N-make.PFV-AOR

Usually, it is quite challenging to isolate vVPE from AE, and, as mentioned before, because of this there is a lot of disagreement about how verb-stranding constructions are derived. The reasoning here should be as follows: if there is a constituent which cannot be deleted independently in the language but in some cases can nevertheless undergo ellipsis along with the internal argument(s), then it must have been silenced by virtue of VPE. The archetypal candidates for this role of non-deletable constituents are adverbs, so the interpretive possibilities of these elements within ellipsis sites should be diagnostic of VPE, see (Goldberg 2005; Simpson et al. 2013; Funakoshi 2016; Manetta 2018; Halder 2021; among many others).

Fortunately enough, in Muira Dargwa, not only are adverbs non-deletable on their own, but also NVs. For reasons that will become clear later, I defer the discussion of the former until §4.1, focusing now on the deletion of NVs. The following paradigm demonstrates that it is permissible to silence either the internal argument and the NV (36a), or the internal argument alone (36b). Crucially, it is impossible to silence the NV independently (36c).

- (36) a. Učitil-li Murad parbat w-arq'-ib, direktur-li = ra
 teacher-ERG Murad.ABS calm M-make.PFV-AOR principal-ERG = ADD
 w-arq'-ib
 M-make.PFV-AOR
 'The teacher calmed Murad, and the principal did too.'

- b. Učitelj-li Murad parbat w-arq'-ib, direktur-li = ra parbat
 teacher-ERG Murad.ABS calm M-make.PFV-AOR principal-ERG = ADD calm
 w-arq'-ib
 M-make.PFV-AOR
 'The teacher calmed Murad, and the principal calmed him, too.'
- c. #Učitelj-li Murad parbat w-arq'-ib, direktur-li = ra Murad
 teacher-ERG Murad.ABS calm M-make.PFV-AOR principal-ERG = ADD Murad.ABS
 w-arq'-ib
 M-make.PFV-AOR
 ('The teacher calmed Murad, and the principal did too.')

Given the impossibility of eliding an NV alone, the analysis (35b) must be incorrect, as it treats the non-pronunciation of an internal argument and an NV as the result of application of two independent operations, thus predicting the availability of NV-drop in (36c), contrary to fact. However, in §3.4, we have encountered examples of sentences where only the NV was left unpronounced, cf. (37). These data, I believe, not only do not pose a problem, but actually provide additional evidence for the existence of ν VPE. An important difference between (36c) and (37) is that in the former example, the internal argument of the ellipsis clause is the same as that of the antecedent clause, while in the latter example they are contrasted. So, the possibility of silencing an NV alone strongly correlates with the presence of a contrastive focus stress on an internal argument, which should allow it to move out of the ellipsis site, surviving silencing. If, however, an internal argument is unable to escape the ellipsis site, as in (36c), where the direct objects of both clauses are the same, its fate is to be unpronounced. These facts, I believe, allow us to conclude that (37) must also be derived via ν VPE.

- (37) Učitelj-li Murad parbat w-arq'-ib, *pro* Rasul_i $\langle t_i \text{ parbat} \rangle$
 teacher-ERG Murad.ABS calm M-make.PFV-AOR Rasul.ABS calm
 w-arq'-b = ak:u
 M-make.PFV-AOR = NEG.COP
 'The teacher calmed Murad down, but RASUL, he didn't.'

Now, one might wonder why the deletion of an NV is not obligatory and how sentences like (36b), in which only the internal argument is missing, are derived. I suggest that, despite the superficial similarity, (36b) might actually be derived via AE, rather than ν VPE. Evidence for yet another ellipsis mechanism is partly negative: I was unable to find any clues pointing to the possibility of an NV incorporating into ν or any other evidence of an NV moving out of the domain of ν VPE.

Besides that, the fact that AE may be responsible for object-drop is confirmed by the presence of sloppy identity readings (38).⁷

- (38) Asijat-li sun-na aba razi r-arq'-ib, Sabijat-li = ra
 Asiyat-ERG self-GEN mother.ABS glad F-make.PFV-AOR Sabiyat-ERG = ADD
 ⟨sun-na aba⟩ razi r-arq'-ib
 self-GEN mother.ABS glad F-make.PFV-AOR
 'Asiyat made her mother happy, and Sabiyat did too.' (Strict – OK; Sloppy – OK)

Simpson et al. (2013) provide data from three South Asian languages – Bangla, Hindi-Urdu and Malayalam – and demonstrate that VVPE and AE can coexist peacefully within the same language. So, there is nothing exceptional about the fact that both processes are at work in Muira Dargwa, although in some contexts it may be extremely difficult to differentiate between the two. For instance, (39) with a simplex predicate may well be derived either by *v*VPE or AE, as the result of applying these processes would be indistinguishable, see also §4.1.

- (39) Muḥammad-li paltar d-erc-ib, Rabada-j ⟨paltar⟩
 Muhammad-ERG clothing.ABS NPL-wash.PFV-AOR Rabadan-ERG clothing.ABS
 ḥa-d-erc-ib
 NEG-NPL-wash.PFV-AOR
 'Muhammad washed the clothes, but Rabadan didn't.'

In this subsection, I have shown that CPRs, and more specifically the impossibility of eliding NVs individually, provide crucial evidence for the existence of *v*VPE in Muira Dargwa. An important lesson from this discussion is that NVs, once their morphosyntactic status has been established, can be productively employed to diagnose (different types of) VPE in other languages as well.⁸ To my knowledge, this has only been done so far for Bangla in (Haldar 2021). Crucially, as can be seen from the paradigm presented in (40)–(43), Bangla demonstrates exactly the same pattern as that observed in Muira Dargwa, thus providing additional support for the reliability of the *NV-drop test*.

⁷ The apparent optionality of the NV deletion is also attested in Persian, where it has enjoyed a lot of attention (Shafiei 2015; Rasekhi 2016; 2018; Ahari 2024). What most of the existing approaches to this issue have in common is that they argue that examples like (36a) and (36b) are derived by virtue of (almost) the same ellipsis operation (*v*VPE or something similar) which targets a rather large constituent, silencing all the material within it in one fell swoop. The main difference between these approaches is how each author achieves the possibility of leaving an NV untouched. For instance, Shafiei (2015) allows an NV to be incorporated into *v*, which results in it leaving the ellipsis site along with the LV; this incorporation is thought to be driven by discourse factors. Ahari (2024) takes a different path and proposes two different VPE-like operations: one targeting the NV and the other not. I believe that my approach, which exploits two ellipsis operations independently attested in the language, is superior for dealing with (at least) Muira Dargwa data.

⁸ I thank an anonymous reviewer for the suggestion to emphasize this fact.

(40) *Bangla* (Halдар 2021: 12)

- A: Moḍ^hu kobiṭa-ṭa muk^host^ho kor-l-o
 Madhu.NOM poem-ACC mouth.residing do-PST-3
 ‘Madhu memorized the poem.’
- B: B^hanu-o kor-l-o
 Bhanu.NOM-too do-PST-3
 ‘Bhanu memorized the poem too.’

(41) *Bangla* (Halдар 2021: 13)

- A: Moḍ^hu kobiṭa-ṭa muk^host^ho kor-l-o
 Madhu.NOM poem-ACC mouth.residing do-PST-3
 ‘Madhu memorized the poem.’
- B: B^hanu-o muk^host^ho kor-l-o
 Bhanu.NOM-too mouth.residing do-PST-3
 ‘Bhanu memorized the poem too.’

(42) *Bangla* (Halдар 2021: 13)

- A: Moḍ^hu kobiṭa-ṭa muk^host^ho kor-l-o
 Madhu.NOM poem-ACC mouth.residing do-PST-3
 ‘Madhu memorized the poem.’
- B: #B^hanu-o kobiṭa-ṭa kor-l-o
 Bhanu.NOM-too poem-ACC do-PST-3
 (‘Bhanu memorized the poem too.’)

(43) *Bangla* (Halдар 2021: 13)

- A: Moḍ^hu kobiṭa-ṭa muk^host^ho kor-et^h-e
 Madhu.NOM poem-ACC mouth.residing do-PFV-PRES.3
 ‘Madhu has memorized the poem.’
- B: Ar B^hanu(-o) gaṇ-ṭa kor-et^h-e
 and Bhanu.NOM(-too) song-ACC do-PFV-PRES.3
 ‘And Bhanu has memorized the SONG.’

4 Triggering ellipsis

It is a general truth that not every XP can undergo ellipsis, and every theory of ellipsis must account for this fact. Most modern theories propose that there are two central requirements that must be satisfied for deletion to occur. These are known as *licensing requirement* and *identity requirement*, thanks to (Lobeck 1995).

(44) a. *Licensing requirement*

Only certain syntactic heads license ellipsis of their complements.

b. *Identity requirement*

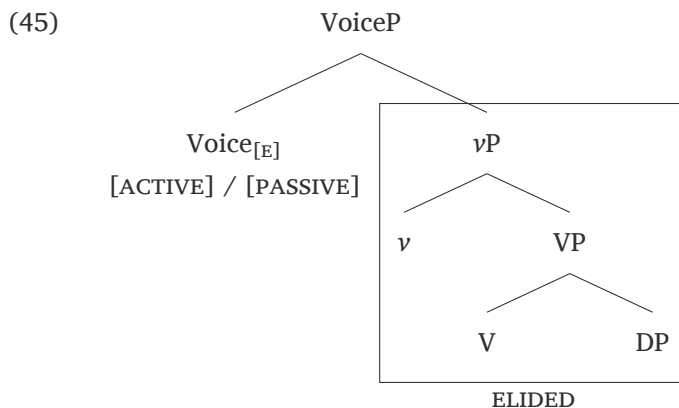
There must be an antecedent of the same syntactic type that matches the elided material.

In this section, I introduce one of the most influential approaches to ellipsis, first formulated in (Merchant 2001), discuss both requirements in more detail and show how Muira Dargwa *v*VPE satisfies them.

4.1 Licensing requirement

The first part of (44) refers to the local, idiosyncratic features of the head that triggers the non-pronunciation of its complement. This requirement can be captured in many different ways. In his now mainstream approach, Merchant (2001) posits a lexical feature [E] (or even a family of features) which instructs PF to block pronunciation of the complement phrase (the semantics of the feature will be given in §4.2). The [E] feature, however, can originate on a fixed number of heads: for English, the list includes at least C, T (or Asp), Voice and D.

In his discussion of English VPE, Merchant (2013c) shows that VPE is the deletion of the complement of Voice; this is schematically represented in (45). Since the Voice head bears the [E] feature and it is thus not located within the ellipsis site, VPE is predicted to tolerate voice mismatches; this will be discussed in more detail in §5.



One more consequence of positing the [E] feature on Voice concerns the possible interpretations of the modifier *again*. Von Stechow (1996) observes that in German, when *wieder* ‘again’ modifies a transitive predicate, it is ambiguous between two readings: *repetitive* and *restitutive*. The ambiguity, as von Stechow claims, is syntactic in nature and it is due to the ability of the adverb to attach in two different structural positions – as an adjunct of either *v*P or VP – where it introduces

a presupposition that the eventuality denoted by the modified constituent has occurred before. Johnson (2004) demonstrates that the same ambiguity also arises in English, see (46).

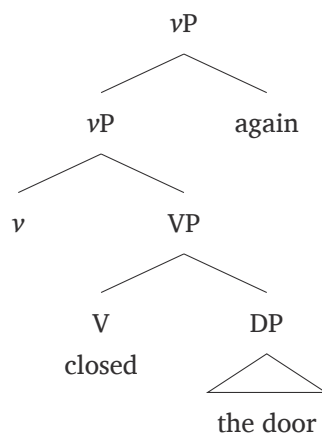
(46) (Johnson 2004: 8)

She closed the door again.

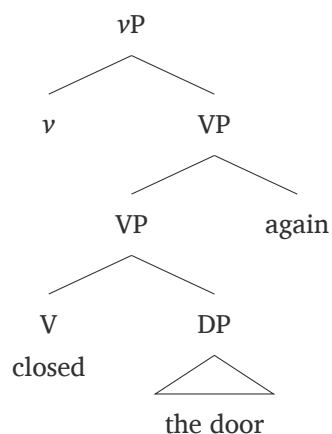
- a. Repetitive: 'She closed the door, and somebody had closed it before.'
- b. Restitutive: 'She closed the door, and it had been in that state before.'

That is, the ambiguity is between the two structures: (47), where *again* is adjoined to the *vP* that denotes the event of the causation of the state denoted by its complement VP, and (48), in which *again* modifies the stative VP that denotes the result state.

(47)



(48)



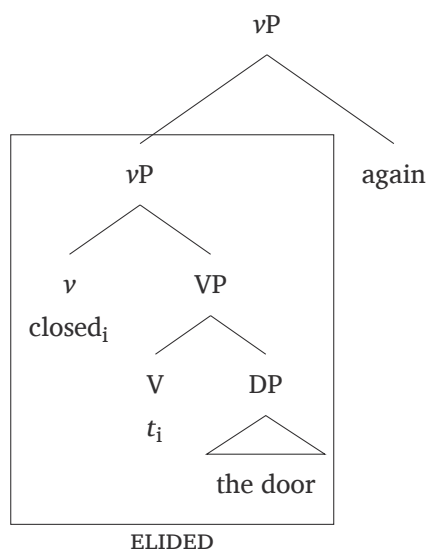
Johnson (2004) further notes that von Stechow's approach to the ambiguity of *again* can be fruitfully implemented to diagnose the constituent that is silenced under ellipsis. It turns out that only repetitive readings are available in English VPE, cf. (49).

(49) (Johnson 2004: 9)

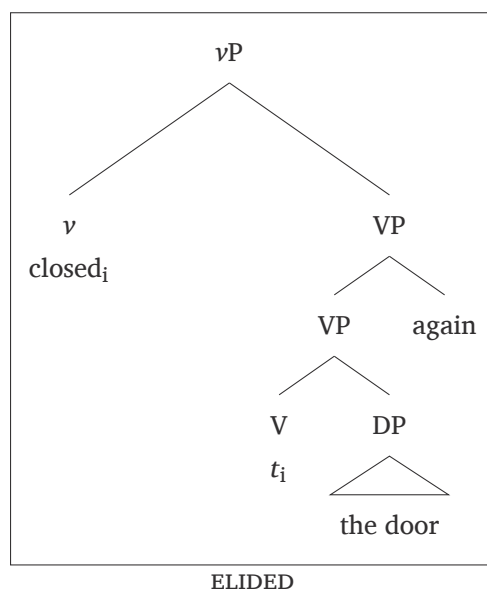
- a. Jane closed the door, and then Maribel did ⟨close the door⟩ again.
- b. *Yesterday the wind blew the door open, and no one closed it. Today, finally, Maribel did ⟨close the door⟩ again.

The contrast is totally expected if VPE deletes an entire vP . The adverb can survive ellipsis only if it merges with the vP (50), giving rise to the repetitive reading (49a). In turn, if *again* is adjoined to the VP, as in (51), it must be deleted, thus making it impossible for the restitutive reading (49b) to arise.

(50)



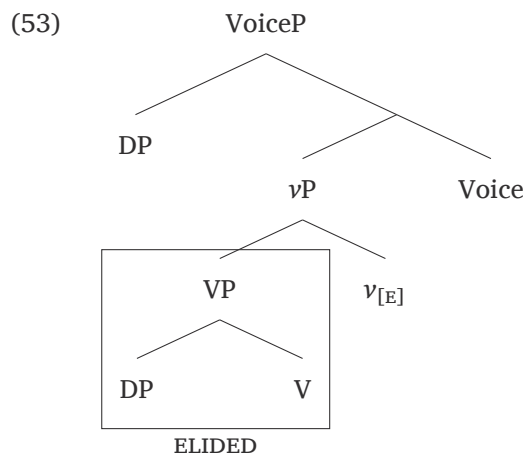
(51)



Turning now to Muira Dargwa, the *again*-test can be employed to adjudicate the size of the phrase that ν VPE targets. Under the current analysis of CPrs, presented in §2.2, the distinction between eventive and stative components is clear. For instance, the NV *hark* ‘open’ of the CPr *hark* AGR-*arq* ‘to open’ denotes the result state of being open, while the LV AGR-*arq* ‘make’ expresses the causing event. When *ʔibrra* ‘again’ is used in the context of ν VPE, quite surprisingly, both readings are permitted: the repetitive reading in (52a) presupposes the previous existence of the opening event, while the restitutive reading in (52b) has the presupposition that there previously existed the state of the door being open.

- (52) a. Murad-li unc:a hark b-arq'-ib, Rasul-li = ra ʔibrra
 Murad-ERG door.ABS open N-make.PFV-AOR Rasul-ERG = ADD again
 ⟨unc:a hark⟩ b-arq'-ib
 door.ABS open N-make.PFV-AOR
 ‘Murad opened the door, and then Rasul did open the door again.’
- b. Daʔ-li unc:a čikawq'-un, čī-li-k'al il hark
 wind-ERG door.ABS close.PFV-AOR who-ERG-INDEF this.ADDR.ABS open
 b-arq'-ib-li ač:^wi. aχirra Rasul-li ʔibrra ⟨unc:a hark⟩
 N-make.PFV-AOR-CVB NEG.COP.PST finally Rasul-ERG again door.ABS open
 b-arq'-ib
 N-make.PFV-AOR
 ‘The wind blew the door shut, no one opened it. Finally, Rasul opened the door again.’

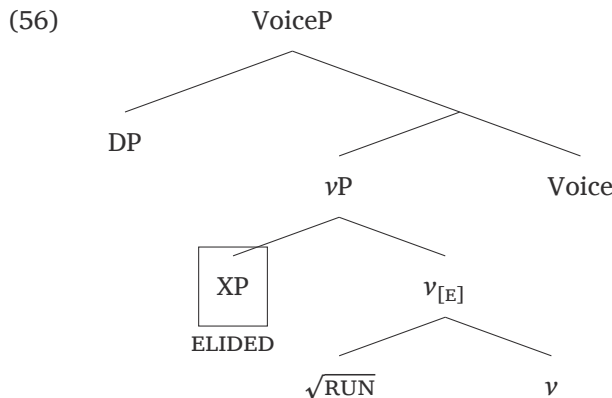
Given the availability of both repetitive and restitutive interpretations of *ʔibrra* ‘again’ in the context of ν VPE, it is reasonable to conclude that ν VPE in Muira Dargwa is smaller than VPE in English. More concretely, I suggest that ν VPE targets the complement of ν . This is schematically represented in (53), where the [E] feature is located on the ν head.



This lower-than-average size of the VPE has a number of consequences. Here, I address only two of them, postponing discussion of others until §5 (some other languages with ellipses of apparently similar size will be discussed in §6). First, it should become evident that the widely employed adjunct-test is of no value in differentiating ν VPE from AE. If, as it is traditionally assumed, low adjuncts are attached to ν P/VoiceP, Muira Dargwa ν VPE should never be able to target them, as they are attached to a projection outside the ellipsis site. AE, in turn, simply deletes individual arguments and cannot target adjuncts. In both (54) and (55), missing adjuncts are not syntactically present, as they cannot be interpreted in the elliptical clause. So, as expected, they are missing completely, and not deleted by ν VPE.

- (54) Rasul-li ma^hhkam-li unc:a hark b-arq'-ib, Asadulla-li
 Rasul-ERG careful-ADV door.ABS open N-make.PFV-AOR Asadulla-ERG
 ĥa-b-arq'-ib
 NEG-N-make.PFV-AOR
 OK: 'Rasul carefully opened the door, but Asadulla didn't open it.'
 #: ('Rasul carefully opened the door, and Asadulla opened it carelessly.')
- (55) Tuχtur-li kumk-li ahik'an ʔa^h Ø-a^hrq'-ib, medsestra-li
 doctor-ERG fast-ADV patient.ABS good M-make.PFV-AOR nurse-ERG
 ĥ-Ø-arq'-ib
 NEG-M-make.PFV-AOR
 OK: 'The doctor quickly cured the patient, but the nurse didn't cure him.'
 #: ('The doctor quickly cured the patient, but the nurse cured him slowly.')

Second, recall the discussion of the structure of CPRs in §2.2. It was suggested that there exist two structural positions where NVs – which correspond to bare roots in the narrow syntax – can be merged. So far, the discussion has been limited to roots that merge with the direct object and describe the end state that is associated with a direct object of a change-of-state event. There are, however, roots that modify the event introduced by ν , for example imposing a manner on an activity event, – these are argued to be attached directly to ν . Consequently, under this view, the proposed analysis predicts an important ellipsis asymmetry: direct object-adjoined NVs must be targeted by ν VPE, while ν -adjoined NVs are expected to be exempt from it. The reason for this is that the [E] feature instructs the non-pronunciation of the complement of the head it is adjoined to. So, if the feature is located on the ν head, as argued here, ν VPE targets the complement of ν . Crucially, unlike direct object-adjoined NVs, ν -adjoined NVs are never contained within the expected ellipsis site and are predicted to be non-elidable. This is schematically represented in (56).



It turns out that this prediction is borne out partially. The status of vVPE in these contexts is subject to some inter-speaker variation. For two consultants, omitting an NV may result only in somewhat degraded acceptability.⁹ For now, I do not have a good explanation for this, so I leave it for further research (though see footnote 9). It should be emphasized, however, that the unacceptability of these examples is not due to the elision of just the NV – in §3.6 it was demonstrated that, under certain circumstances, it is indeed possible for vVPE to elide only the NV.

- (57) *Darħa taʰh Ø-uq-un, rursi = ra r-uq-un
 child.ABS jump M-move.PFV-AOR girl.ABS = ADD F-move.PFV-AOR
 ‘The boy jumped, and the girl did too.’

- (58) ??Murad ducʰ Ø-uq-un, Musa = ra Ø-uq-un
 Murad-ABS run M-move.PFV-AOR Musa.ABS = ADD M-move.PFV-AOR
 ‘Murad ran, and Musa did too.’

- (59) ??Darħa ʔimč w-ikʰ-u-li saj, rursi = ra r-ikʰ-u-li
 child.ABS sneeze M-say.IPFV-PROG-CVB COP.M girl.ABS = ADD F-say.IPFV-PROG-CVB
 sa < r > i
 < F > COP
 ‘The boy sneezes, and so does the girl.’

4.2 Identity requirement

In the previous subsection, it was established that the head which hosts the [E] feature and licenses ellipsis is *v*. It was also noted that the [E] feature not only gives PF the instruction to not pronounce the complement of the head it is adjoined to, but it is also used to capture the identity requirement,

⁹ I have also consulted three native speakers of Icari Dargwa, South Dargwa language that before recently was spoken in a village of Icari (Sumbatova & Mutalov 2000). All of them strongly reject similar stimuli where NVs of activity CPrs are missing.

introduced in (44). Obviously, the identity to an antecedent may be of different nature, see (Merchant 2013b) for an overview of existing approaches. In his seminal work, Merchant (2001) argues for a semantic identity requirement that he calls E-GIVENness, see (61) for the definition.¹⁰ The [E]-feature, as can be seen in (60), is endowed with the presupposition that the complement must be E-GIVEN. Thus, [E] plays a dual role, acting as a mediator between PF and LF.

(60) (Merchant 2001: 61)

$$\llbracket E \rrbracket = \lambda p: p \text{ is E-GIVEN}$$

(61) (Merchant 2018: 259)

An expression E counts as E-GIVEN iff E has a salient antecedent (expressed or implied) A and, modulo λ -type-shifting,

- a. $\llbracket A \rrbracket \in \llbracket E \rrbracket^f$
- b. $\llbracket E \rrbracket \in \llbracket A \rrbracket^f$

To illustrate how it all works, the interpretation of CPRs must first be elaborated. As discussed in §2.2, I follow (Marantz 2013) and much subsequent work by taking roots to be syntactic and semantic modifiers describing a particular event. With respect to DPs, I assume that some DPs are interpreted purely as entities, while others can be interpreted as events. A DP can be eventive either by virtue of its inherent meaning or because it is coerced into being eventive, specifically a state. This coercion is the result of the application of the rule (62), see (Marantz 2009a;b).

(62) $\llbracket DP \rrbracket \rightarrow \text{STATE}(\llbracket DP \rrbracket) = \lambda e.\text{state}(e, DP)$

As for the semantic composition, besides strictly compositional rules, such as *Functional Application* and *Predicate Modification* (Heim & Kratzer 1998), the elements can also be combined by the non-compositional rule of *Event Identification* (Kratzer 1996), defined in (63).

(63) *Event Identification*

If α is a branching node, $\{\beta, \gamma\}$ is the set of α 's daughters, where $\llbracket \beta \rrbracket$ is in $D_{\langle e, \langle s, t \rangle \rangle}$ and $\llbracket \gamma \rrbracket$ is in $D_{\langle s, t \rangle}$, then $\llbracket \alpha \rrbracket = \lambda x_e.\lambda e_s.\llbracket \beta \rrbracket(x)(e) \wedge \llbracket \gamma \rrbracket(e)$.

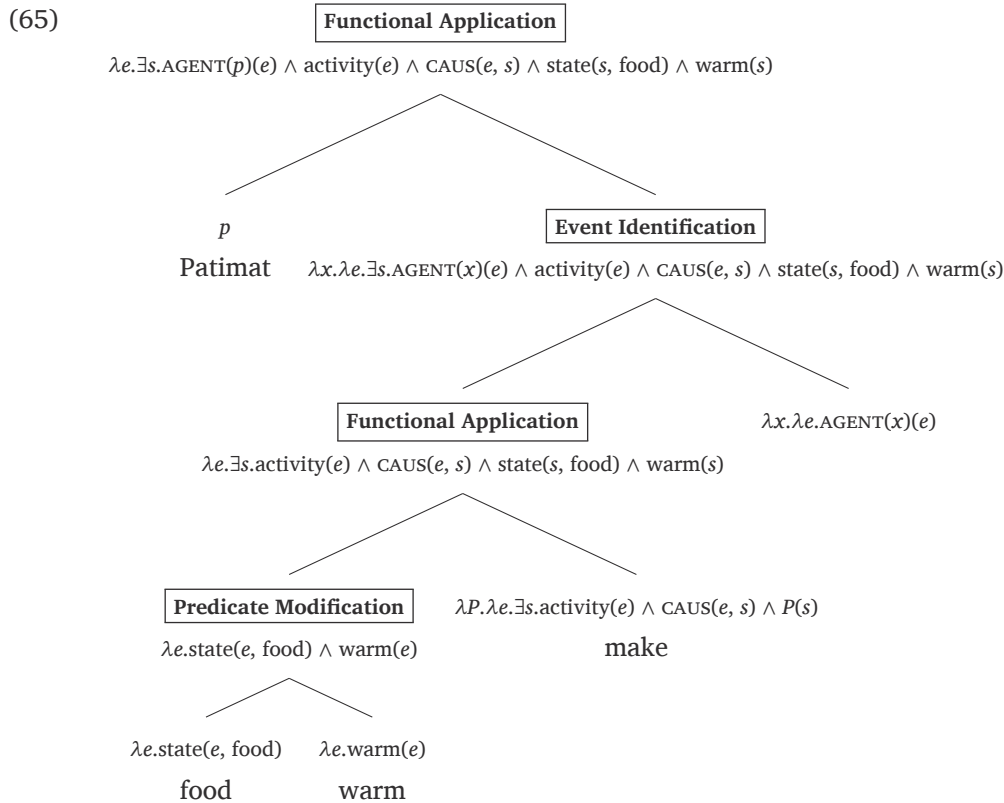
¹⁰ The original formulation of E-GIVENness was as in (i). However, for reasons having to do with certain problems arising because of the mutual entailment requirement (Murphy 2016; Merchant 2018), in (Merchant 2018) the condition (i) was reformulated as in (61). Specifically, the $\exists$ -type shifting operation was changed to a λ -type shifting operation, and the mutual entailment condition to a mutual inclusion condition.

- (i) An expression E counts as E-GIVEN iff E has a salient antecedent (expressed or implied) A and, modulo $\exists$ -type-shifting,
 - a. A entails F-clo(E) and
 - b. E entails F-clo(A)

(Merchant 2001: 26)

Taking this into account, but nevertheless simplifying somewhat, the denotation of (64) with a change-of-state CPr can be represented as in (65), where P is a variable over eventive predicates, e a variable over events, and x a variable over individuals.

- (64) Pat'imat-li χ ureg wana b-arq'-ib
 Patimat-ERG food.ABS warm N-make.PFV-AOR
 'Patimat warmed up the food.'



Now we are in a position to examine how the identity requirement is checked under ν VPE. Let us look at the baseline example (3), repeated here as (66).

- (66) Pat'imat-li χ ureg wana b-arq'-ib, Ashura-li=ra $\langle \chi$ ureg wana
 Patimat-ERG food.ABS warm N-make.PFV-AOR Ashura-ERG=ADD food.ABS warm
 b-arq'-ib
 N-make.PFV-AOR
 'Patimat warmed up the food, and Ashura did too.'

In order for the mutual inclusion conditions to apply, the focus marked elements undergo λ -type shifting operation. In the case of (66), however, there are no such elements. The

first part of the definition in (61) requires that the denotation of the target include the denotation of the antecedent. (67) shows that the denotations are identical, so the requirement is trivially satisfied.

- (67) a. $\llbracket [\chi_{ureg} wana]_A \rrbracket = \lambda e.state(e, food) \wedge warm(e) \in$
 b. $\llbracket [\chi_{ureg} wana]_E \rrbracket^f = \lambda e.state(e, food) \wedge warm(e)$

The second part of (61) reverses the relationship between the target and antecedent. Now, the denotation of the former must be included in the denotation of the latter. Obviously, as the denotations are identical, this is also the case.

- (68) a. $\llbracket [\chi_{ureg} wana]_E \rrbracket = \lambda e.state(e, food) \wedge warm(e) \in$
 b. $\llbracket [\chi_{ureg} wana]_A \rrbracket^f = \lambda e.state(e, food) \wedge warm(e)$

Both parts of the identity requirement are satisfied, and the ellipsis is licit. In these examples, F-closure doesn't play any significant role, as nothing in the target or antecedent clauses is F-marked. Let us now turn to examples where it has some weight.

Sentences that resemble pseudogapping, discussed in §3.6, pass the identity check in a similar way. Consider (69). In both (70) and (71), the contrastive focus marked internal argument in F-closed clause is abstracted over by λ -type-shifting, thus making the clause a superset of the denotation of the other clause. So, the ellipsis here is also licit.

- (69) Učitol-li Musa parvat w-arq'-ib, *pro* Rasul_i $\langle t_i \text{ parvat} \rangle$
 teacher-ERG Musa.ABS calm M-make.PFV-AOR Rasul.ABS calm
 w-arq'-b = ak:u
 M-make.PFV-AOR = NEG.COP
 'The teacher calmed Musa down, but RASUL, he didn't.'

- (70) a. $\llbracket [musa \text{ parvat}]_A \rrbracket = \lambda e.state(e, Musa) \wedge calm(e) \in$
 b. $\llbracket [rasul \text{ parvat}]_E \rrbracket^f = \lambda x.\lambda e.state(e, x) \wedge calm(e)$
 (71) a. $\llbracket [rasul \text{ parvat}]_A \rrbracket = \lambda e.state(e, Rasul) \wedge calm(e) \in$
 b. $\llbracket [musa \text{ parvat}]_E \rrbracket^f = \lambda x.\lambda e.state(e, x) \wedge calm(e)$

In this subsection, I have demonstrated that due to the rather small amount of the deleted material, the identity requirement can be easier to satisfy. In the majority of cases, the calculation is trivial. With this, however, there emerge a number of consequences. I now proceed to discuss some of them.

5 Argument structure alternations

This section discusses additional consequences that follow from the proposed analysis. All of them are related to argument structure and its flexibility: inchoative-causative alternations (§5.1), causativization (§5.2), and antipassivization (§5.3). I demonstrate that the (im)possibility of mismatches between antecedent and elided material can be accounted for within current analysis without additional stipulations. Then, in §5.4, I discuss the ramifications for some other theories of ellipsis.

5.1 *v* and ellipsis: inchoative-causative alternations

In §4.1, it was noted that, according to Merchant (2013c), VPE in English targets the *v*P, leaving outside the Voice head, which is responsible for hosting the voice features. As (72) shows, this correctly predicts the grammaticality of voice mismatches between the target and antecedent clauses, see (Poppels & Kehler 2019; 2025) and references therein for a more in-depth discussion and important qualifications.

(72) (Merchant 2013c: 79)

The system can be used by anyone who wants to ⟨use it⟩.

For ellipses that target constituents larger than the *v*P, voice mismatches result in ungrammaticality; this is illustrated in (73) for sluicing.

(73) (Merchant 2013c: 81)

*Joe was murdered, but we don't know who ⟨murdered Joe⟩.

Merchant, following a long tradition in literature on argument structure, associates causative semantics with the *v* head (Chomsky 1995; Hale & Keyser 2002; Pylkkänen 2008), which is immediately below Voice. So, another important prediction of his analysis is the impossibility of transitivity alternations under any type of ellipsis in English, even VPE. Example (74) demonstrates that the prediction is borne out (see, however, §6). Indeed, this observation is not new, but Merchant seems to be the first to give a principled explanation for the ban on such ellipses.

(74) (Sag 1976: 160)

*Bill melted the copper vase, and the magnesium vase did ⟨melt⟩, too.

Returning now to Muira Dargwa, if the analysis proposed here is on the right track, *v*VPE is expected to tolerate transitivity alternations, as the operation does not target the *v* head, which is responsible for transitivity and eventive semantics. Let us look at examples (75) and (76). In both cases, CPRs denote the same warming up event. However, one important difference between the target and antecedent clauses in these examples is that one of the CPRs introduces an agent

argument, while the other one does not. This difference, as argued in §2.2, is determined by which Voice and *v* heads are selected. In the case of inchoative CPrs, Voice is specified so as not to introduce an external argument, and *v* is such that it simply denotes a change-of-state event, no external force is presupposed. In turn, causative CPrs are formed with the help of a thematic Voice head that must have a DP in its specifier, and a causative *v* which implies the presence of an agent. Despite this mismatch, ellipsis in both (75) and (76) is licit. So, the initial prediction that mismatches in the content of *v* should not play a role in *v*VPE is borne out.

(75) *Inchoative antecedent, causative target*

- A: Šin wana d-Ø-ib-li = w?
 water.ABS warm NPL-become.PFV-AOR-CVB = PQ
 ‘Has the water warmed up?’
- B: Nu-ni ⟨šin wana⟩ d-arq’-ib = ra
 I-ERG water.ABS warm NPL-make.PFV-AOR = 1
 ‘I warmed it up.’

(76) *Causative antecedent, inchoative target*

- A: Hu-ni šin wana d-arq’-at:i = w?
 you-ERG water.ABS warm NPL-make.PFV-AOR.2SG = PQ
 ‘Have you warmed up the water?’
- B: Il-tri_i ⟨t_i wana⟩ d-Ø-ib
 this.ADDR.ABS-PL warm NPL-become.PFV-AOR
 ‘It warmed up (on its own).’

It should be noted that the internal argument of an inchoative CPr is considered to move from the *v*P to the surface subject position, leaving a copy inside the ellipsis site. Under the assumption that A-movement reconstructs, as it has been argued to do for reasons of scope (May 1977), the satisfaction of the identity requirement appears to be as trivial as in the example (66), discussed in the previous section. That is, the identity check between the target and its antecedent in (75) and (76) is exactly as in (67) and (68), modulo lexical entries.

5.2 Going up: causativization

Causativization is the only productive process of valency change in Dargwa languages. In all languages of the group, both morphological and periphrastic causative constructions are attested. Here, I will discuss only the former, since the latter have not been thoroughly investigated yet and, it seems, will not affect the current discussion in any meaningful way.

Morphological causative is formed by adding the affix *-iq* directly to the verbal base. The process of causativization is highly productive – causatives are easily formed from verbs with

all types of argument structure: intransitive (both unaccusative and unergative), transitive, ditransitive and affective. The subject of the causative construction is always in the ergative case. The arguments of the causativized verb are expressed exactly the way they are in the non-causative correlate, with the only exception of the external argument of a causativized transitive/ditransitive predicate – its case marking is changed from ergative to inter-lative. Example (77) illustrates how the construction is formed on the transitive base.

(77) a. *Transitive*

Musa-li niʔ d-uč:-u-li saj
 Musa-ERG milk.ABS NPL-drink.IPFV-PROG-CVB COP.M
 ‘Musa is drinking milk.’

b. *Causativized transitive*

At:a-li Musa-ci niʔ d-uč:-iq-u-li saj
 father-ERG Musa-INTER milk.ABS NPL-drink.IPFV-CAUS-PROG-CVB COP.M
 ‘Father is making Musa drink milk.’

Since (Pylkkänen 2008), morphological causative formation is regarded as adding a dedicated causative head on top of either *vP* or *VoiceP*. So, it is predicted that the exact type of Muira Dargwa causative is irrelevant, because in both scenarios *vVPE* is too small to target the relevant fragment of the structure.¹¹ As can be seen in (78), LVs, being outside the ellipsis site, need not match – it is possible to have a causative LV in the antecedent clause and a causativized inchoative LV in the target clause, or vice versa.

- (78) Pat’imat-li χureg wana b-arq’-ib, Ashura-li = ra ⟨χureg wana⟩
 Patimat-ERG food.ABS warm N-make.PFV-AOR Ashura-ERG = ADD food.ABS warm
 b-Ø-iq-ib
 N-become.PFV-CAUS-AOR
 ‘Patimat warmed up the food, and Ashura did too.’

Moreover, as was the case with inchoative-causative pairs, discussed in the previous subsection, transitivity mismatches are also tolerated, see (79) and (80). Thus, the proposed analysis makes correct predictions here as well.

(79) *Intransitive antecedent, causativized intransitive target*

A: Šin wana d-Ø-ib-li = w?
 water.ABS warm NPL-become.PFV-AOR-CVB = PQ
 ‘Has the water warmed up?’

¹¹ See (Kalyakin 2025a) for some evidence that in a related lect of Kaytag Dargwa morphological causative is Voice-selecting.

- B: Nu-ni ⟨šin wana⟩ d-Ø-iq-ib = ra
 I-ERG water.ABS warm NPL-become.PFV-CAUS-AOR = ADD
 ‘I warmed it up.’

(80) *Transitive antecedent, causativized transitive target*

- A: Rasul-li šin wana d-arq’-ib-li = w?
 Rasul-ERG water.ABS warm NPL-make.PFV-AOR-CVB = PQ
 ‘Has Rasul warmed up the water?’
 B: Musa-li (il-i-ci) ⟨šin wana⟩ d-arq’-iq-ib
 Musa-ERG this.ADDR-OBL-INTER water.ABS warm NPL-make.PFV-CAUS-AOR
 ‘Musa made him do it.’

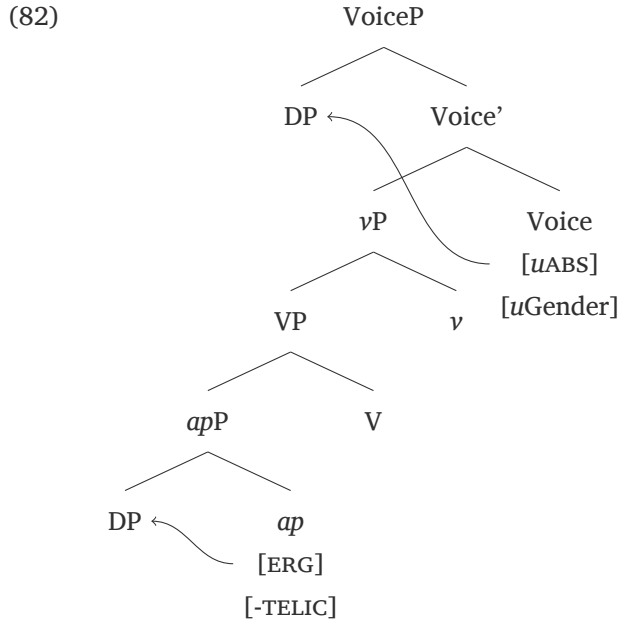
5.3 Going down: antipassivization

Dargwa languages are characterized by the presence of an antipassive construction, which is quite rare in Nakh-Daghestanian languages, see (Comrie et al. 2021) for an overview. In this construction, the logical object of a transitive predicate is realized not as a direct object, but as an oblique argument, adjunct, or is not expressed at all (Polinsky 2017). In Dargwa languages, antipassive is quite peculiar in that it does not have any special morphological marking. Semantically, it is limited to imperfective aspect and is used to express some kind of habituality or iterativity. Example (81) shows an ergative construction and its corresponding antipassive correlate. Superficially, (81a) and (81b) differ in case marking of the arguments and agreement on the lexical verb – these are the main morphosyntactic reflexes of the antipassive construction.

- (81) a. *Ergative construction*
 Učeniku-n-a-li žuz-i d-uč’-u-li sa i
 student-PL-OBL.PL-ERG book.ABS-PL NPL-read.IPFV-PROG-CVB <HPL> COP
 ‘Students are reading books.’
 b. *Antipassive construction*
 Učeniku-ni žuz-Ø-a-li b-uč’-u-li sa i
 student.ABS-PL book-PL-OBL.PL-ERG HPL-read.IPFV-PROG-CVB <HPL> COP
 ‘Students are engaged in reading books.’

Trying to capture these and some other properties of the antipassive construction, Kalyakin (2024) proposes the structure (82). In this structure, the internal argument is licensed by a special functional head *ap* (*freely Merging Case-assigning head*, Yuan 2018), which is located rather low – just above the internal argument. This head is aspectual in nature and is endowed with the ability to assign inherent ergative case. The external argument is licensed by the intransitive

Voice head, so it receives structural absolutive case. This derivation is argued to explain the lack of agreement with the internal argument and the fact that the construction is limited to the imperfective aspect.



The semantics of *ap* have yet to be studied. Though, Kalyakin (2024), following Wharram (2003) and Deal (2008), suggests that the preliminary denotation may be along the lines of (83). As can be observed, the operator mediates the composition of a property-type object with a verb, providing existential closure of the former. This approach correctly predicts obligatory narrow scope and non-specific (*de dicto*) interpretation of the internal argument.

$$(83) \quad \llbracket ap \rrbracket = \lambda P_{\langle e, t \rangle}. \lambda Q_{\langle e, \langle s, t \rangle \rangle}. \lambda e. \exists x. P(x) \wedge Q(e, x)$$

Given the low structural position of the *ap* head, it is worth investigating how antipassivization interacts with ν VPE. First, witness (84), which demonstrates that the CPr *dawk* AGR-*irq'* ‘to repair’ may be used in the antipassive construction.

- (84) Bašir qur-r-a-li dawk w-irq'-u-li saj
 Bashir.ABS house-PL-OBL.PL-ERG repaired M-make.IPFV-PROG-CVB COP.M
 ‘Bashir is engaged in repairing houses.’

Turning now to ν VPE, ellipsis is infelicitous if an antipassive construction is anteceded by an ergative construction (85), or vice versa. This seems to be expected, as the semantic contribution of the antipassive operator should make the denotations of the two clauses sufficiently different.

(85) *Transitive antecedent, antipassive target*

#Bašir-li qur-ri dawk b-irq'-u-li saj, Musa = ra
 Bashir-ERG house.ABS-PL repaired NPL-make.IPFV-PROG-CVB COP.M Musa.ABS = ADD
 ⟨qur-r-a-li dawk⟩ w-irq'-u-li saj
 house-PL-OBL.PL-ERG repaired M-make.IPFV-PROG-CVB COP.M
 ('Bashir is engaged in repairing houses, and so is Musa.')

This, however, is not the full story. It turns out that *v*VPE is prohibited even in those cases where both clauses contain antipassive predicates, cf. (86).

(86) *Antipassive antecedent, antipassive target*

#Bašir qur-r-a-li dawk w-irq'-u-li saj,
 Bashir.ABS house-PL-OBL.PL-ERG repaired M-make.IPFV-PROG-CVB COP.M
 Musa = ra ⟨qur-r-a-li dawk⟩ w-irq'-u-li saj
 Musa.ABS = ADD house-PL-OBL.PL-ERG repaired M-make.IPFV-PROG-CVB COP.M
 ('Bashir is engaged in repairing houses, and so is Musa.')

No matter how mysterious this situation may seem at first glance, I would like to claim that this is also to be expected. Recall the discussion of unergatives in §2.2 and §4.1. It was suggested that in the case of these predicates, roots impose a manner on an activity event and are attached directly to *v*. With this, NVs of these CPrs cannot be targeted for deletion by *v*VPE. In turn, antipassives are usually considered to also denote activity events (Polinsky 2017) – it seems to be the case for Muira Dargwa as well. Given this parallel between unergative and antipassive predicates, I propose that under antipassivization roots that are usually attached to direct objects are coerced into being exceptionally merged as modifiers of the *v* head. Consequently, as in the case of regular *v*-attaching NVs, *v*VPE is unable to target the NV of an antipassive CPr. If this is on the right track, the unacceptability of (85) must be not due to the lack of identity between the antecedent and ellipsis clauses, as initially suggested, but rather because the NVs in antipassives resist non-pronunciation for structural reasons – they are too high to be deleted.¹²

5.4 *v*VPE and current theories of ellipsis

Now, having explored how *v*VPE interacts with argument structure alternations, it is worthwhile to discuss what existing theories of ellipsis have to say about the observed non-trivial empirical picture. The analysis proposed above was based on Merchant's (2001; 2013c) assumptions about the crucial role of the size of parallelism domains. Specifically, it has been argued that ellipsis tolerates mismatches between the ellipsis clause and the antecedent clause, provided that the

¹² I thank an anonymous reviewer for pointing out the relevance of extending this logic to (85).

relevant head is located outside the constituent targeted for deletion. In Muira Dargwa, due to the lower structural position of the [E] feature, *v*VPE allows for a greater number of mismatches than, for example, VPE in English.

Other approaches that do not appeal to the size of the elided constituent seem to face serious problems in explaining the availability of the indicated mismatches in *v*VPE. For some theories that posit only semantic identity based on entailment relations (Dalrymple et al. 1991) and for inference-based (Culicover & Jackendoff 2005) theories the puzzle is why transitivity mismatches should, in principle, be allowed, since inchoative and causative clauses are not mutually entailing (the denotation of the former is a proper subset of the denotation of the latter) and do not allow for the relevant inferences.

A similar problem holds for discourse-based theories of mismatches under ellipsis (Kehler 2002). According to Kehler's (2002) approach, the (un)availability of certain mismatches depends on discourse coherence relations between the clauses that contain the antecedent and elided material. In his discussion of voice mismatches under VPE in English, Kehler shows that two families of discourse relations are relevant: the *Resemblance* family of relations and the *Cause-Effect* family. Resemblance relations allow sentences of a discourse to be connected by emphasizing "commonalities and contrasts among corresponding sets of entities and relations" (Kehler 2002: 15). In turn, utterances that stand in a Cause-Effect relation require "a path of implication connecting a pair of propositions" in the two sentences (Kehler 2002: 20).

Kehler argues that when there is a voice mismatch in VPE, sentences where there is a Cause-Effect relation between antecedent and ellipsis site are licit, while sentences where there is a Resemblance relation are illicit. The contrast can be observed in (87).

- (87) a. *Cause-Effect* (Kehler 2002: 53)
 This problem was to have been looked into, but obviously nobody did.
 b. *Resemblance* (Kehler 2002: 56)
 *This problem was looked into by John, and Bob did too.

Despite considerable success in explaining the (un)availability of voice mismatches in VPE, Kehler's theory faces serious challenges when it comes to explaining the behavior of Muira Dargwa *v*VPE. According to Kehler, sentence (88) produces ill-formed discourse: the expected Resemblance relation cannot be established because the agent of the causative clause has no correspondence in the inchoative clause. Contrary to expectations, all Muira Dargwa speakers I have consulted find (88) fully acceptable.

- (88) Pat'imat-li χureg wana b-arq'-ib, šin_i ⟨t_i wana⟩ d-Ø-ib
 Patimat-ERG food.ABS warm N-make.PFV-AOR water.ABS warm NPL-become.PFV-AOR
 'Patimat warmed up the food, and the water warmed up too.'

The acceptability of *v*VPE in (88) and the contrast between (87a) and (87b), however, is completely expected for Sailor's (2014) structural approach to featural mismatches in VPE. Building on (Merchant 2013c), Sailor contends that the most immediate factor dictating the (un)availability of mismatches is syntax, not discourse coherence. Based on this, he considers VPE to be a non-uniform operation: in environments that allow voice mismatches, the Voice head is outside the ellipsis site (*low VPE*); in environments that prohibit voice mismatches, Voice is inside the ellipsis site (*high VPE*). Thus, the contrast between (87a) and (87b) boils down to the type of VPE employed: *low VPE* in (87a) and *high VPE* in (87b). For this approach, therefore, the acceptability of (88) is anticipated, since *v* – the head responsible for transitivity and eventive semantics – is outside the ellipsis site, and thus free to differ featurally from its antecedent. So, I suggest that the data from Muira Dargwa *v*VPE provides novel evidence in support of Sailor's (2014) structural approach, further expanding the typology of VPE with *extra low VPE* – a subtype of VPE that targets the complement of *v*.

Finally, it is important to emphasize that this line of analysis makes a number of predictions about possible featural mismatches in ellipses of different size. Merchant (2013c) claims that there is no language that would be an inverse of English allowing voice mismatches in TP, but not VP, ellipses. We can extend this logic and suggest that it is impossible to imagine an inverse of Muira Dargwa – a language that would not tolerate transitivity mismatches in *v*VPE, but allow for them in, for example, sluicing. Additionally, it is predicted that if an ellipsis operation allows for transitivity mismatches between the antecedent and ellipsis clauses, it should also allow for voice mismatches. Unfortunately, this prediction is hard to test on Muira Dargwa data, as the language does not have rich voice morphology. So, it is worth identifying some other languages with a similar ellipsis process and investigating the patterns of available featural mismatches.

6 *v*VPE crosslinguistically

Before concluding, I would like to discuss some other languages in which *v*VPE has been argued to be present. The reason for doing so is to place Muira Dargwa within a broader context and to explore whether there exist other ways of getting a *v*VPE-like string, just like it was argued for both VPE (Sailor 2014) and VVPE (Lipták 2012; Gribanova 2013). So, I believe it would be useful to come up with an initial typology of the construction and identify some available parameters of variation.

In §1, it was noted that *v*VPE has also been attested in Persian and first described in (Toosarvandani 2009), though see (Shafiei 2015; Rasekhi 2016; 2018; Ahari 2024) for critical discussion and alternative proposals. As can be seen from example (89), in Persian, just as in Muira Dargwa, both the internal argument and the NV can be silent, while the LV remains overtly expressed.

(89) (Toosarvandani 2009: 2)

Sohrāb piranā-ro otu na-zad vali Rostam ⟨piranā-ra otu⟩ zad
 Sohrab shirt.PL-ACC iron NEG-hit.PST.3SG but Rostam shirt.PL-ACC iron hit.PST.3SG
 ‘Sohrab didn’t iron the shirts, but Rostam did.’

The parallels between Persian and Muira Dargwa do not end there. Toosarvandani employs the *again*-test, discussed in §4.1, and concludes that Persian *v*VPE is as small as its Muira Dargwa relative, i.e. it also targets the complement of *v*. The reason for this is that both repetitive and restitutive readings are readily available in the context of *v*VPE (below I provide an example only for the latter reading).

(90) (Toosarvandani 2009: 16)

Dishab āshpazxune pāk bud. Leylā omad kasif-esh kard.
 last.night kitchen clean was Leila come.PST.3SG dirty-it do-PST.3SG
 Kasi na-raft pāk-esh bo-kone. Emshab maxām dobāre
 nobody NEG-go.PST.3SG clean-it SUBJ-do.3SG tonight want.PRES.1SG again
 ⟨āshpazxune pāk⟩ bo-konam
 kitchen-ACC clean SUBJ-do.3SG
 ‘Last night, the kitchen was clean. Leila came and dirtied it. Nobody went to clean it.
 Tonight, I will clean it again.’

The two languages, however, differ in at least one important respect. Namely, Persian *v*VPE does not tolerate argument structure alternations, cf. (91).

(91) (Toosarvandani 2009: 22)

A: Lebāsā xoshk shodan?
 clothes dry become.PART.3PL
 ‘Have the clothes dried yet?’
 B: #Na, vali Rostam alān raft ⟨lebāsā-ro xoshk⟩ bo-kone.
 no but Rostam now go.PST.3SG clothes-ACC dry SUBJ-do.3SG
 (‘No, but Rostam just went to dry them.’)

Even more than that, the language seems to completely prohibit any ellipsis where the LVs of the antecedent and target clauses do not match. This restriction on the identity of LVs is strikingly similar to an effect that Goldberg (2005) observed in VVPE languages and labeled as Verbal Identity Requirement (VIR).

(92) *Verbal Identity Requirement* (Goldberg 2005: 186)

The antecedent- and target-clause main Vs of VP Ellipsis must be identical, minimally, in their root and derivational morphology.

Despite the fact that Toosarvandani explicitly rejects the possibility of VIR being active in Persian, and instead offers a non-uniform treatment of the ban on LV alternations, data from Muira Dargwa presented in §5 may suggest that, ultimately, something similar to VIR could be behind the ban on LV alternations in Persian. So, just as “VVPE languages” differ in whether they comply with VIR or not, “vVPE languages” may also either respect or disregard VIR.

Besides Muira Dargwa and Persian, vVPE has also been encountered in Hindi-Urdu (Manetta 2018; 2019) and Bangla (Halдар 2021). Here, I will limit my discussion to Bangla only, because, according to the available data, the workings of vVPE in these languages are quite similar. First, witness (93), which is superficially identical to the analogous examples from Muira Dargwa and Persian.

(93) (Halдар 2021: 12)

- A: Moḍ^hu kobaṭa-ṭa muk^host^ho kor-l-o
 Madhu.NOM poem-ACC mouth.residing do-PST-3
 ‘Madhu memorized the poem.’
- B: B^hanu-o kor-l-o
 Bhanu.NOM-too do-PST-3
 ‘Bhanu memorized the poem too.’

However, as Halдар (2021) shows, the structure underlying the vVPE string in (93) is actually different. Once again, the *again*-test plays a crucial role here. In (94), the contrast between the acceptable repetitive reading of ‘again’ (94a) and the unacceptable restitutive reading of the modifier (94b) allows us to conclude that ellipsis in Bangla targets a bigger portion of the clause – the vP, according to Halдар.

(94) (Halдар 2021: 20)

- a. ɒmol ʃɒkal-e g^hɒr-ṭa porɪʃ^hkar kor-et^hi-l-o. Pɒre ami abar
 Amol.NOM morning-LOC room-ACC clean do-PFV-PST-3 later I.NOM again
 kor-l-am
 do-PST-1
 ‘Amol had cleaned the room in the morning. Later, I did again.’
- b. #fiɑɑ-ɛ ɖɒrɕɑ-ṭa k^hul-e gæ-l-o. Keɪ bɒŋɖ^ho kor-l-o n-a, ʈai
 wind-ERG door-NOM open-GER go-PST-3 anyone closed do-PST-3 NEG-IPFV so
 ami abar kor-l-am
 I.NOM again do-PST-1
 (‘The wind blew the door open. No one closed it, so I did again.’)

Given the bigger size of the ellipsis, Bangla, unlike Muira Dargwa, should allow missing adjuncts to be interpreted in the deleted constituent. This turns out to be right even in the case of simple

predicates (95), for which the distinction between VVPE and AE is now not obscured, see the discussion in §3.6.

(95) (Haldar 2021: 5)

- A: Moḍ^hu ḍu-g^hḍṇṭa ḍ^hore æk-ta kobiṭa lik^h-l-o
 Madhu.NOM two-hour for one-CLSFR poem write-PST-3
 ‘Madhu wrote a poem for two hours.’
- B: B^hanu-o lik^h-l-o
 Bhanu.NOM-too write-PST-3
 ‘Bhanu wrote a poem (for two hours) too.’

So, the derivation of Bangla *v*VPE is almost the same as that of VVPE, that is, the LV must evacuate the ellipsis site before the deletion occurs. Nevertheless, it should be emphasized that *v*VPE in languages like Bangla may additionally inform the debate about whether crosslinguistic VVPE exists or not. Landau (2020a;b) argues that this type of ellipsis should be unavailable due to locality constraints on the interactions between head movement and ellipsis. CPrs, in addition to adjuncts, may pose a problem for such a view, because, as discussed in §3.6, there seems to exist no such operation as NV-drop, so missing NVs must be missing due to an operation that, unlike AE, deletes a rather large portion of a clause in one fell swoop.

So far, the discussion of *v*VPE has been limited to CPrs exclusively. However, this need not be so – one can expect to encounter cases of *v*VPE in some other contexts as well. In other words, CPrs are not prerequisites for *v*VPE. It turns out that this intuition is confirmed. Silk (2025) argues that British *do* construction, exemplified in (96), instantiates *v*VPE.

(96) (Silk 2025: 1)

Alice will bring snacks, and Bert will do, too.

In order to differentiate between two analyses, – one where *do* realizes Voice, and one where it is the realization of *v* – Silk presents experimental evidence from voice mismatches and argument structure mismatches that supports treating British *do* ellipsis as eliding the complement of *v*. The reason for this conclusion is that this ellipsis tolerates both types of mismatches, see (97) and (98). Recall that VPE in English, which was discussed in §5, allows the former, while completely prohibiting the latter.

(97) (Silk 2025: 7)

The matter should be looked into, but I doubt anyone will do before the meeting.

(98) (Silk 2025: 12)

The travellers tried to close the suitcase, but it wouldn't *(do).

So, British *do* ellipsis appears to delete the same constituent as Muira Dargwa ν VPE, namely the complement of ν . Given the presence of the ellipsis construction, which is argued for in this paper, in such a well-studied language as British English, it may be the case that ν VPE has been overlooked yet in some other languages as well.

7 Conclusion and outlook

Though VPE in its diverse varieties has been encountered and investigated in a number of typologically unrelated languages, almost nothing was known about its existence, let alone its workings in the languages of the Nakh-Daghestanian family. In this paper, I have argued that Muira Dargwa exhibits an under-described type of VPE, labeled here as ν -Stranding Verb Phrase Ellipsis – an ellipsis process which targets an internal argument along with the non-verbal element of a complex predicate, leaving the light verb stranded. Similarly to VPE in English and VVPE in head-raising languages, ν VPE obeys the same well known requirements (Lobeck 1995). This ellipsis process, however, is peculiar in that the ellipsis-triggering [E] feature is located rather low in the syntactic structure – namely, it is adjoined to the ν head. This fact has a number of important consequences, distinguishing Muira Dargwa ν VPE from its bigger relatives. I have addressed the ambiguity of the modifier ‘again’ between repetitive and restitutive readings and discussed why some argument structure mismatches (inchoative-causative alternations, causativization) are tolerated, while others (antipassivization) are not. The heads responsible for the former processes are argued to be external to the antecedent and elided phrases, hence not part of the material that needs to be relevantly identical for the ellipsis to be licit. In turn, antipassivization is considered to resist ellipsis due to a coercion process that forces roots that are usually attached to direct objects to be attached as modifiers of the ν head. In these scenarios, as in the case of unergative predicates, NVs turn out to be outside the ellipsis site. If this explanation is on the right track, these data may be taken to support the neo-constructionalist approaches to argument structure (Marantz 2013), according to which roots are acategorical elements that are devoid of syntactic information and, importantly, can in principle attach in different structural positions. And more generally, I have shown that the data presented in this paper strongly argue in favor of structural approaches to ellipsis phenomena (Merchant 2013c; Sailor 2014) which attribute the variable effect of featural mismatches across different types of ellipsis to the size of the elided constituent.

Given the fact that ν is stranded because the constituent that is deleted is smaller than that hosting the verbal element, ν VPE in Muira Dargwa extends the typology of ellipsis constructions in the languages of the world. The existence of such an ellipsis, where verb-stranding is achieved not via head movement, is perhaps predicted, given the long-standing assumption that ellipsis applies to phrases of any category as long as its licensing requirements are met. However, languages in which ν VPE is not that small, and the construction’s derivation is more similar to that of VVPE,

i.e. the LV evacuates the ellipsis site before the deletion occurs, are also attested. Languages belonging to the latter group might inform the debate about whether crosslinguistic VVPE exists or not, since, as I argue here, there is no such thing as NV-drop, i.e. an operation that elides the NV of a CPr. Further work will show whether ν VPE also exists in some other Nakh-Daghestanian languages and how rare such ellipsis is crosslinguistically.

Finally, with this work, there emerges a number of questions. For instance, it still remains to be investigated why such ellipsis may exist in a language and why a language may employ two different ellipsis processes – VPE and AE – in addition to the extensive use of null *pro*-forms. A potential answer to the first question may appeal to the requirement that the ellipsis-licensing head – ν in case of ν VPE – must be overtly and independently realized. This, in a sense, is similar to the traditional approaches to VPE (Lobeck 1995; Aelbrecht 2010). As for the second question, it may well be the case that *pro*-drop will prove to be a much more limited phenomenon, with the majority of putative cases of null arguments being successfully derived via AE (Duguine 2014; Mizuno 2025).

Abbreviations

The examples in this paper broadly follow the Leipzig Glossing Rules. The abbreviations are as follows: 1 = First person, 2 = Second person, 3 = Third person, ABS = absolutive, ACC = accusative, ADD = additive, ADDR = addressee, ADV = adverb, AOR = aorist, ATR = attributive, CAUS = causative, CLSFR = classifier, COP = copula, CVB = converb, ERG = ergative, F = feminine, GEN = genitive, GER = gerund, H = human, INDEF = indefinite, IPFV = imperfective, IQ = indirect question, LOC = locative, LV = light verb, M = masculine, N = neuter, NEG = negative, NOM = nominative, OBL = oblique, PART = participle, PFV = perfective, PL = plural, PQ = polar question, PRES = present; PROG = progressive, PST = past, SG = singular, SP = speaker, SUBJ = subject, TH = thematic element, UP = upward.

Funding information

This research has been supported by the Russian Science Foundation, project 25-18-00222 “Control and Raising in the languages of Eurasia” realized at Pushkin State Russian Language Institute.

Acknowledgements

I am deeply grateful to my Kalkni friends and consultants Abdullah Abdullaev, Asiyat Abdullaeva, Malikat Abdullaeva, and Zainab Ibieva for their hospitality, invaluable input and acceptability judgments. I am also thankful to three anonymous reviewers and the editor David Erschler for their very useful comments and suggestions on earlier versions of this paper. Finally, I would like to thank Margatita Ivanova, Lada Pasko, Dmitry Ganenkov, Daniar Kasenov, and Nina Sumbatova for extensive discussions of various points of this paper, which significantly improved the text.

Competing interests

The author has no competing interests to declare.

References

- Aelbrecht, Lobke. 2010. *The syntactic licensing of ellipsis*. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/la.149>
- Ahari, Komeil. 2024. Head-adjunction is not enough: Deriving Farsi complex predicates under consistent leftward movements. In *Proceedings of Third North American Conference in Iranian Linguistics*. Available at: <https://ling.auf.net/lingbuzz/007618>
- Alexiadou, Artemis. 2014. Roots don't take complements. *Theoretical Linguistics* 40(3–4). DOI: <https://doi.org/10.1515/tl-2014-0012>

- Baltin, Mark. 2012. Deletion versus pro-forms: An overly simple dichotomy? *Natural Language & Linguistic Theory* 30(2). 381–423. DOI: <https://doi.org/10.1007/s11049-011-9157-x>
- Chomsky, Noam. 1995. *The minimalist program*. Cambridge, MA: MIT Press.
- Comrie, Bernard & Forker, Diana & Khalilova, Zaira & van den Berg, Helma. 2021. Antipassives in Nakh-Daghestanian languages: Exploring the margins of a construction. In Janic, Katarzyna & Witzlack-Makarevich, Alena (eds.), *Antipassive: Typology, diachrony, and related constructions*, 515–548. John Benjamins Publishing Company. DOI: <https://doi.org/10.1075/tsl.130.16com>
- Culicover, Peter & Jackendoff, Ray. 2005. *Simpler syntax*. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/acprof:oso/9780199271092.001.0001>
- Dalrymple, Mary & Shieber, Stuart M. & Pereira, Fernando C. N. 1991. Ellipsis and higher-order unification. *Linguistics and Philosophy* 14(4). 399–452. DOI: <https://doi.org/10.1007/BF00630923>
- De Belder, Marijke & van Craenenbroeck, Jeroen. 2015. How to merge a root. *Linguistic Inquiry* 46(4). 625–655. DOI: https://doi.org/10.1162/ling_a_00196
- Deal, Amy Rose. 2008. Property-type objects and modal embedding. In *Proceedings of Sinn und Bedeutung*, 92–106. Oslo: ILOS.
- Duguine, Maia. 2014. Argument ellipsis: A unitary approach to pro-drop. *The Linguistic Review* 31(3–4). 515–549. DOI: <https://doi.org/10.1515/tlr-2014-0010>
- Erschler, David. 2021. Ellipsis in languages of the Caucasus. In Polinsky, Maria (ed.), *The Oxford handbook of languages of the Caucasus*, 943–972. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oxfordhb/9780190690694.013.26>
- Folli, Raffaella & Harley, Heidi & Karimi, Simin. 2005. Determinants of event type in Persian complex predicates. *Lingua* 115(10). 1365–1401. DOI: <https://doi.org/10.1016/j.lingua.2004.06.002>
- Forker, Diana. 2021. Information structure in languages of the Caucasus. In Polinsky, Maria (ed.), *The Oxford handbook of languages of the Caucasus*, 973–1002. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oxfordhb/9780190690694.013.28>
- Funakoshi, Kenshi. 2016. Verb-stranding verb phrase ellipsis in Japanese. *Journal of East Asian Linguistics* 25(2). 113–142. DOI: <https://doi.org/10.1007/s10831-016-9143-8>
- Ganenkoy, Dmitry & Maisak, Timur. 2021. Nakh-Dagestanian languages. In Polinsky, Maria (ed.), *The Oxford handbook of languages of the Caucasus*, 86–145. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oxfordhb/9780190690694.013.4>
- Goldberg, Lotus. 2005. *Verb-stranding VP ellipsis: A cross-linguistic study*. McGill University dissertation.
- Gribanova, Vera. 2013. Verb-stranding verb phrase ellipsis and the structure of the Russian verbal complex. *Natural Language & Linguistic Theory* 31(1). 91–136. DOI: <https://doi.org/10.1007/s11049-012-9183-3>
- Halдар, Shrayana. 2021. Bengali verb-stranding VP ellipsis and ellipsis identity conditions. Unpublished manuscript. Available at: <https://ling.auf.net/lingbuzz/006247>

- Hale, Kenneth L. & Keyser, Samuel Jay. 2002. *Prolegomenon to a theory of argument structure*. Cambridge, MA: MIT Press. DOI: <https://doi.org/10.7551/mitpress/5634.001.0001>
- Hankamer, Jorge & Sag, Ivan. 1976. Deep and surface anaphora. *Linguistic Inquiry* 7. 391–428.
- Harley, Heidi. 2013. External arguments and the Mirror Principle: On the distinctness of Voice and v. *Lingua* 125. 34–57. DOI: <https://doi.org/10.1016/j.lingua.2012.09.010>
- Heim, Irene & Kratzer, Angelika. 1998. *Semantics in generative grammar*. Oxford: Blackwell.
- Hoji, Hajime. 1998. Null object and sloppy identity in Japanese. *Linguistic Inquiry* 29(1). 127–152. DOI: <https://doi.org/10.1162/002438998553680>
- Irwin, Patricia. 2012. *Unaccusativity at the interfaces*. New York University dissertation.
- Jayaseelan, Karattuparambil. 1990. Incomplete VP deletion and gapping. *Linguistic Analysis* 20. 64–81.
- Johnson, Kyle. 2001. What VP ellipsis can do, and what it can't, but not why. In Baltin, Mark & Collins, Chris (eds.), *The handbook of contemporary syntactic theory*, 439–479. Wiley. DOI: <https://doi.org/10.1002/9780470756416.ch14>
- Johnson, Kyle. 2004. How to be quiet. In *Proceedings from the Annual Meeting of the Chicago Linguistic Society*, 1–20.
- Johnson, Kyle. 2009. Gapping is not (VP-)ellipsis. *Linguistic Inquiry* 40. 289–328. DOI: <https://doi.org/10.1162/ling.2009.40.2.289>
- Kalyakin, Ivan. 2023. Complex predicates in Muira Dargwa. Unpublished manuscript.
- Kalyakin, Ivan. 2024. Antipassive construction in Muira Dargwa. Talk given at *First Eurasian Congress of Linguistics, Institute of Linguistics, RAS*, December 9–13.
- Kalyakin, Ivan. 2025a. Morphological causative in Kaytag Dargwa and the (im)possibility of causative recursion. Manuscript submitted for publication.
- Kalyakin, Ivan. 2025b. Noun-verb complex predicates in Dargwa languages. Talk given at *Caucasian Linguistics: Recent Advances and Emerging Topics, Laboratoire Dynamique du Langage, Lyon*, June 11–13.
- Kasenov, Daniar & Rudnev, Pavel. 2025. Experimental evidence for argument ellipsis as the only derivation of Russian verb-stranding constructions. Unpublished manuscript. Available at: <https://ling.auf.net/lingbuzz/008258>
- Kehler, Andrew. 2002. *Coherence, reference, and the theory of grammar*. Stanford: CSLI Publications.
- Koryakov, Yuri. 2021. Dargin languages and their classification. In Maisak, Timur & Sumbatova, Nina & Testelet, Yakov (eds.), *Durkhasi khazna: A collection of articles for the 60th anniversary of R. O. Mutalov*, 139–154. Moscow: Buki-Vedi.
- Kratzer, Angelika. 1996. Severing the external argument from its verb. In Haegeman, Liliane & Maling, Joan & McCloskey, James & Rooryck, Johan & Zaring, Laurie (eds.), *Phrase structure and the lexicon*, 109–137. Dordrecht: Springer Netherlands. DOI: https://doi.org/10.1007/978-94-015-8617-7_5

- Landau, Idan. 2020a. Constraining head-stranding ellipsis. *Linguistic Inquiry* 51(2). 281–318. DOI: https://doi.org/10.1162/ling_a_00347
- Landau, Idan. 2020b. On the nonexistence of verb-stranding VP-ellipsis. *Linguistic Inquiry* 51(2). 341–365. DOI: https://doi.org/10.1162/ling_a_00346
- Levinson, Lisa. 2014. The ontology of roots and verbs. In Alexiadou, Artemis & Borer, Hagit & Schäfer, Florian (eds.), *The syntax of roots and the roots of syntax*, 208–229. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/acprof:oso/9780199665266.003.0010>
- Lipták, Anikó. 2012. V-stranding ellipsis and verbal identity: The role of polarity focus. *Linguistics in the Netherlands* 29. 82–96. DOI: <https://doi.org/10.1075/avt.29.07lip>
- Lobeck, Anne. 1995. *Ellipsis: Functional heads, licensing, and identification*. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oso/9780195091816.001.0001>
- Manetta, Emily Walker. 2018. The structure of complex predicates in Hindi-Urdu: Evidence from verb-phrase ellipsis. In Sharma, Ghanshyam & Bhatt, Rajesh (eds.), *Trends in Hindi Linguistics*, 47–84. De Gruyter. DOI: <https://doi.org/10.1515/9783110610796-003>
- Manetta, Emily Walker. 2019. Verb-phrase ellipsis and complex predicates in Hindi-Urdu. *Natural Language & Linguistic Theory* 37(3). 915–953. DOI: <https://doi.org/10.1007/s11049-018-9429-9>
- Manetta, Emily Walker. 2021. Verb-second and the verb-stranding verb phrase ellipsis debate. *Glossa: a journal of general linguistics* 6(1). DOI: <https://doi.org/10.16995/glossa.5704>
- Marantz, Alec. 1997. No escape from syntax: Don't try morphological analysis in the privacy of your own lexicon. In *University of Pennsylvania Working Papers in Linguistics* 4, 201–225.
- Marantz, Alec. 2009a. Resultatives and *re*-resultatives: Direct objects may construct events by themselves. Paper presented at the *University of Pennsylvania Linguistics Speaker Series*, Philadelphia, February.
- Marantz, Alec. 2009b. Roots, *re*-, and affected agents: Can roots pull the agent under little *v*? Talk given at *Roots workshop*, Universität Stuttgart, June 10–13.
- Marantz, Alec. 2013. Verbal argument structure: Events and participants. *Lingua* 130. 152–168. DOI: <https://doi.org/10.1016/j.lingua.2012.10.012>
- May, Robert. 1977. *The grammar of quantification*. Massachusetts Institute of Technology dissertation.
- Megerdooonian, Karine. 2011. The status of the nominal in Persian complex predicates. *Natural Language & Linguistic Theory* 30(1). 179–216. DOI: <https://doi.org/10.1007/s11049-011-9146-0>
- Merchant, Jason. 2001. *The syntax of silence: Sluicing, islands, and the theory of ellipsis*. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oso/9780199243730.001.0001>
- Merchant, Jason. 2005. Fragments and ellipsis. *Linguistics and Philosophy* 27(6). 661–738. DOI: <https://doi.org/10.1007/s10988-005-7378-3>
- Merchant, Jason. 2013a. Diagnosing ellipsis. In Cheng, Lisa Lai-Shen & Corver, Norbert (eds.), *Diagnosing syntax*, 537–542. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/acprof:oso/9780199602490.003.0026>

- Merchant, Jason. 2013b. Ellipsis: A survey of analytical approaches. In van Craenenbroeck, Jeroen & Temmerman, Tanja (eds.), *A handbook of ellipsis*. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oxfordhb/9780198712398.013.2>
- Merchant, Jason. 2013c. Voice and ellipsis. *Linguistic Inquiry* 44(1). 77–108. DOI: https://doi.org/10.1162/ling_a_00120
- Merchant, Jason. 2018. Verb-stranding predicate ellipsis in Greek, implicit arguments, and ellipsis-internal focus. In Mikkelsen, Line & Rudin, Deniz & Sasaki, Kelsey (eds.), *A reasonable way to proceed: Essays in honor of Jim McCloskey*, 229–269. UC Santa Cruz: UC eScholarship Repository.
- Merchant, Jason. 2019. Roots don't select, categorial heads do: Lexical-selection of PPs may vary by category. *The Linguistic Review* 36(3). 325–341. DOI: <https://doi.org/10.1515/tlr-2019-2020>
- Mizuno, Teruyuki. 2025. Argument ellipsis as topic deletion. *Natural Language & Linguistic Theory*. DOI: <https://doi.org/10.1007/s11049-024-09645-1>
- Murphy, Andrew. 2016. Subset relations in ellipsis licensing. *Glossa: a journal of general linguistics* 1(1). DOI: <https://doi.org/10.5334/gjgl.61>
- Mutalov, Rasul. 2021. The classification of the Dargin languages and dialects. *Sociolinguistics* 7(3). DOI: <https://doi.org/10.37892/2713-2951-3-7-8-25>
- Otani, Kazuyo & Whitman, John. 1991. V-raising and VP-ellipsis. *Linguistic Inquiry*, 345–358.
- Polinsky, Maria. 2017. Antipassive. In Coon, Jessica (ed.), *The Oxford handbook of ergativity*, 308–331. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oxfordhb/9780198739371.013.13>
- Poppels, Till & Kehler, Andrew. 2019. Reconsidering asymmetries in voice-mismatched VP-ellipsis. *Glossa: a journal of general linguistics* 4(1). DOI: <https://doi.org/10.5334/gjgl.738>
- Poppels, Till & Kehler, Andrew. 2025. Novel cases of sluicing with mismatched antecedents: Theoretical consequences. *Glossa: a journal of general linguistics* 10(1). DOI: <https://doi.org/10.16995/glossa.15307>
- Pylkkänen, Liina. 2008. *Introducing arguments*. Cambridge, MA: MIT Press.
- Rasekhi, Vanideh. 2016. Missing objects in Persian. *Cahier de Studia Iranica* 58. 157–174.
- Rasekhi, Vanideh. 2018. *Ellipsis and information structure: Evidence from Persian*. Stony Brook University dissertation.
- Sag, Ivan. 1976. *Deletion and logical form*. Massachusetts Institute of Technology dissertation.
- Sailor, Craig. 2014. *The variables of VP ellipsis*. University of California, Los Angeles dissertation.
- Schachter, Paul. 1977. Does she or doesn't she? *Linguistic Inquiry* 8. 763–767.
- Schuyler, Tami. 2001. *Wh-movement out of the site of VP ellipsis*. University of California, Santa Cruz dissertation.
- Shafiei, Nazila. 2015. Ellipsis in Persian complex predicates: VVPE or something else? In *Proceedings of the 2015 Annual Conference of the Canadian Linguistic Association*.
- Silk, Alex. 2025. British do: v-stranding ellipsis? Unpublished manuscript. Available at: <https://ling.auf.net/lingbuzz/009099>

- Simpson, Andrew. 2023. In defense of verb-stranding VP ellipsis. *Syntax* 26(4). 431–448. DOI: <https://doi.org/10.1111/synt.12261>
- Simpson, Andrew & Choudhury, Arunima & Menon, Mythili. 2013. Argument ellipsis and the licensing of covert nominals in Bangla, Hindi and Malayalam. *Lingua* 134. 103–128. DOI: <https://doi.org/10.1016/j.lingua.2013.06.007>
- Sumbatova, Nina. 2021. Dargwa. In Polinsky, Maria (ed.), *The Oxford handbook of languages of the Caucasus*, 146–200. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oxfordhb/9780190690694.013.12>
- Sumbatova, Nina & Mutalov, Rasul. 2000. *A grammar of Icarl Dargwa*. München: LINCOM Europa.
- Tabak, Ömer & Demirok, Ömer. 2024. Revisiting complex predicate formation in Turkish. In *Proceedings of the Workshop on Turkic and Languages in Contact with Turkic*, vol. 9, 106–120. DOI: <https://doi.org/10.3765/p5dctm28>
- Toosarvandani, Maziar. 2009. Ellipsis in Farsi complex predicates. *Syntax* 12(1). 60–92. DOI: <https://doi.org/10.1111/j.1467-9612.2008.01115.x>
- Von Stechow, Armin. 1996. The different readings of *wieder* ‘again’: A structural account. *Journal of Semantics* 13(2). 87–138. DOI: <https://doi.org/10.1093/jos/13.2.87>
- Weir, Andrew. 2014. *Fragments and clausal ellipsis*. Amherst: University of Massachusetts dissertation.
- Wharram, Douglas. 2003. *On the interpretation of (un)certain indefinites in Inuktitut and related languages*. University of Connecticut dissertation.
- Yuan, Michelle. 2018. *Dimensions of ergativity in Inuit: Theory and microvariation*. Massachusetts Institute of Technology dissertation.

